CoEval: A Self-Evaluating LLM Ensemble Framework for Scalable, Attribute-Controlled Benchmark Generation

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Abstract

Static benchmarks are non-extensible, prohibitively expensive to construct for novel tasks, and -- critically -- do not reflect the specific data distributions, quality criteria, or edge cases of particular deployment contexts. LLM-as-judge approaches scale annotation throughput but introduce systematic biases: positional preference reverses 20--27% of pairwise rankings, and single-judge correlations with human ratings fall below rho = 0.40 on open-ended generation tasks. No existing method allows practitioners to automatically construct task-specific, attribute-controlled evaluation benchmarks with calibrated multi-judge scoring grounded in explicit rubric criteria. CoEval addresses this gap by introducing a teacher/student/judge paradigm for self-evaluating LLM ensembles. Teacher LLMs define the evaluation space: they enumerate task-relevant quality attributes, author rubric criteria for each attribute, and generate reference (prompt, response) pairs through attribute-stratified sampling that guarantees coverage of the full quality dimension space -- including low-frequency conditions that uncontrolled generation systematically omits. Student LLMs produce candidate responses under evaluation. Judge LLMs independently score each response against the rubric, and OLS calibration corrects for verbosity and positional biases inherent in individual judges, enabling reliable ensemble aggregation without human annotation at any stage. We validate CoEval through medium-benchmark-v1, an experiment spanning four NLP tasks (text summarization, code explanation, email composition, data interpretation) and five models (GPT-4o-mini, GPT-3.5-Turbo, Qwen2.5-0.5B/1.5B, SmolLM2-1.7B). The pipeline produces 7,978 valid evaluations at a total API cost of $5.89 -- between 135 and 1,354 times cheaper than equivalent human annotation -- completing in 12.8 hours with no manual intervention. Pairwise judge agreement spans kappa = 0.003 (SmolLM2-1.7B, negligible) to kappa = 0.422 (GPT-3.5-Turbo x GPT-4o-mini, moderate), demonstrating that ensemble composition is a first-order variable in evaluation reliability. CoEval is fully open-source and configurable via a declarative YAML interface.[*]

Keywords: LLM evaluation, benchmark generation, ensemble scoring, LLM-as-judge, attribute-controlled generation, inter-rater agreement

1 Introduction

The rapid proliferation of large language models across high-stakes deployment contexts has created a structural demand for rigorous, domain-adaptive evaluation infrastructure that current methods cannot satisfy. Human-curated benchmarks -- BIG-Bench (Srivastava et al., 2022), HELM (Liang et al., 2022), MMLU (Hendrycks et al., 2020) -- provide reproducible, ground-truth measurements at a cost of 0.10--1.00 per annotated item, requiring months of expert effort to construct and yielding static, domain-locked artifacts with no adaptation mechanism for novel deployment tasks. LLM-as-judge systems -- G-Eval (Liu et al., 2023), MT-Bench (Zheng et al., 2023), PandaLM (Wang et al., 2023) -- scale annotation throughput but introduce systematic biases: positional preference reverses 20--27% of pairwise rankings when candidate presentation order is swapped, verbosity bias inflates scores for longer outputs regardless of quality, and single-judge Spearman correlations with human ratings fall below rho = 0.40 on open-ended generation tasks. Rubric-based systems such as Prometheus-2 (Kim et al., 2024) partially address the bias problem through structured criteria and pairwise ranking, but require user-supplied rubrics and pre-sourced evaluation items, providing no automated rubric generation, no attribute-stratified benchmark construction, and no multi-judge agreement monitoring.

Standard benchmarks further fail practitioners along a second, more fundamental axis: they evaluate LLMs on fixed, generic tasks that bear no necessary relationship to the specific data distributions, quality criteria, or edge cases of any particular deployment context. A practitioner evaluating a summarization model for medical literature cannot apply MMLU; an organization evaluating customer support tone cannot use BIG-Bench. A legal document review

tool requires evaluation items that reflect the distribution of clauses, jurisdiction-specific terminology, and failure modes encountered in actual legal practice -- not the balanced task mixture of a general-purpose benchmark. No existing framework allows practitioners to automatically construct a tailored, attribute-controlled benchmark for their specific use case: one that systematically samples the quality dimensions relevant to their domain, authors rubric criteria grounded in their quality standards, and produces calibrated multi-judge scores without requiring human annotation expertise. This specific-use-case construction gap is the primary motivation for CoEval.

CoEval addresses this gap through a teacher/student/judge paradigm for self-evaluating LLM ensembles. The paradigm assigns LLMs to three functionally distinct roles. Teacher LLMs construct the evaluation space: given a task description and a set of user-specified quality attributes (e.g., factual accuracy, conciseness, domain terminology adherence), teachers enumerate measurable scoring criteria, author rubrics that operationalize each criterion, and generate reference (prompt, response) pairs that instantiate the evaluation space. Student LLMs produce candidate responses to the teacher-generated prompts, their outputs constituting the primary object of evaluation. Judge LLMs independently score each student response against the rubric criteria, with no access to other judges' scores during scoring. Because judges operate independently and their disagreements are measured quantitatively, systematic biases present in any individual judge -- positional, verbosity, or self-enhancement biases -- become observable and correctable rather than silently propagating into final scores. This closed-loop design -- where the same pool of LLMs can serve as teachers, students, and judges within a single configured run -- requires no human annotation at any stage of routine operation.

CoEval is also a concrete, open-source software system that implements this paradigm in a configurable, fault-tolerant pipeline. The system takes a declarative YAML configuration file as its sole input, through which practitioners specify the task, the quality attributes of interest, and which models serve in each role. Internally, CoEval automates attribute-stratified sampling -- ensuring that generated evaluation items cover the full attribute space, including low-frequency conditions that uncontrolled generation systematically omits -- and automated rubric construction, in which teacher LLMs generate scoring criteria from the attribute specification rather than requiring practitioners to author rubrics manually. Inter-judge agreement is quantified by Strict Pairwise Agreement (SPA), Weighted Pairwise Agreement (WPA), and Cohen's kappa. OLS calibration corrects for inter-judge verbosity and positional biases by regressing each judge's raw scores against available ground-truth signals, enabling reliable ensemble aggregation. Automated filters J *(judge), T* (teacher), and D* (datapoint) identify and exclude low-reliability contributors before score aggregation, transforming inter-judge diagnostics from descriptive statistics into actionable quality controls. The pipeline supports full artifact checkpointing and phase-level resume for fault tolerance across long-running experiments.

These two design choices -- attribute-stratified sampling and OLS-calibrated multi-judge scoring -- directly address the two gaps identified above. Because CoEval samples evaluation items from a user-defined attribute space rather than from a fixed, generic task distribution, the resulting benchmark reflects the practitioner's specific deployment context: the attributes, the quality criteria, and the edge-case distribution are all under practitioner control. A medical summarization practitioner specifies attributes such as clinical accuracy, information completeness, and contraindication salience; CoEval generates evaluation items that cover each attribute and scores student responses against rubric criteria grounded in those attributes, not in the generic quality dimensions of a standard benchmark. Because CoEval employs multiple independent judges with OLS-corrected aggregation, the reliability gap of single-judge LLM evaluation is addressed systematically: judge bias is measured, biased judges are filtered by J*, and the remaining ensemble's agreement is reported as a first-class experimental result rather than an implicit assumption.

We validate both the paradigm and the system through medium-benchmark-v1, an experiment spanning four NLP tasks (text summarization, code explanation, email composition, data interpretation) and five models (GPT-4o-mini, GPT-3.5-Turbo, Qwen2.5-0.5B, Qwen2.5-1.5B, SmolLM2-1.7B). The pipeline produces 7,978 valid scored evaluations at a total API cost of 5.89 USD -- between 135 and 1,354 times cheaper than equivalent human annotation at 0.10--$1.00 per item -- completing in 12.8 hours with no manual intervention across all pipeline phases. Judge reliability spans from kappa = 0.422 (moderate agreement, Landis and Koch, 1977) for the GPT-3.5-Turbo/GPT-4o-mini pair to kappa = 0.003 (negligible agreement) for SmolLM2-1.7B, confirming that ensemble composition is a first-order variable in automated evaluation quality. A non-obvious finding from teacher discrimination analysis reveals that SmolLM2-1.7B, the weakest model by capability, ranks first on V1 discrimination (V1 = 0.0046), suggesting that prompt diversity induced by weaker generators outweighs generation quality as a determinant of benchmark informativeness.

This paper makes the following contributions:

Conceptual contributions:

- **Teacher/student/judge paradigm.** A formally defined role separation that organizes LLM ensembles into generators, respondents, and evaluators, enabling closed-loop benchmark construction and evaluation without human annotation.
- **Teacher discrimination metrics.** Automated metrics V1 (inter-datapoint score variance on a squared scale), S2 = sqrt(V1) (interpretability rescaling to score units), and R3 (range: maximum minus minimum student mean score) quantify how well each teacher LLM induces meaningful performance differences between student models without human labels, supporting data-driven teacher selection.

System contributions:

- **Unified configurable pipeline.** To our knowledge, CoEval is the first framework to integrate attribute-controlled benchmark generation, automated rubric construction, multi-judge ensemble scoring, OLS calibration, and inter-judge agreement monitoring in a single open-source, YAML-configurable pipeline. The specific functional gaps it fills relative to the four closest prior systems (Prometheus-2, ARES, G-Eval, Self-Instruct) are tabulated in Table IC1 (Section 3).
- **Robust ensemble filters.** Judge filter J*, teacher filter T*, and datapoint filter D *identify and exclude low-reliability contributors before score aggregation. Empirical evidence: J* exclusion of SmolLM2-1.7B (kappa = 0.003) raises ensemble kappa from a projected 0.152 to a measured 0.422 (full ablation planned as future work).
- **Controlled attribute sampling.** Stratified sampling over user-specified target and nuanced attributes ensures systematic coverage of the deployment input distribution, including evaluation-critical low-frequency conditions that uncontrolled generation omits.

Empirical contributions:

- **medium-benchmark-v1 validation with full artifact release.** All experimental configurations, generated datasets, evaluation JSONL logs, fitted OLS coefficients, and analysis reports are publicly released at https://github.com/ApartsinProjects/CoEval. The full analysis pipeline can be re-executed from stored logs in under 5 minutes with no additional API expenditure using coeval analyze --run-dir Runs/medium-benchmark-v1/.

Section 2 surveys prior work on benchmark construction, LLM-as-judge evaluation, and synthetic data generation that CoEval builds upon and distinguishes itself from. Section 3 presents the framework design and methodology in full detail, including the attribute specification schema, rubric construction procedure, and ensemble filtering criteria. Section 4 reports experimental results across tasks, models, and cost dimensions. Section 5 analyzes ensemble failure modes, negative results, and cost efficiency. Section 6 states limitations and future directions. Section 7 concludes.

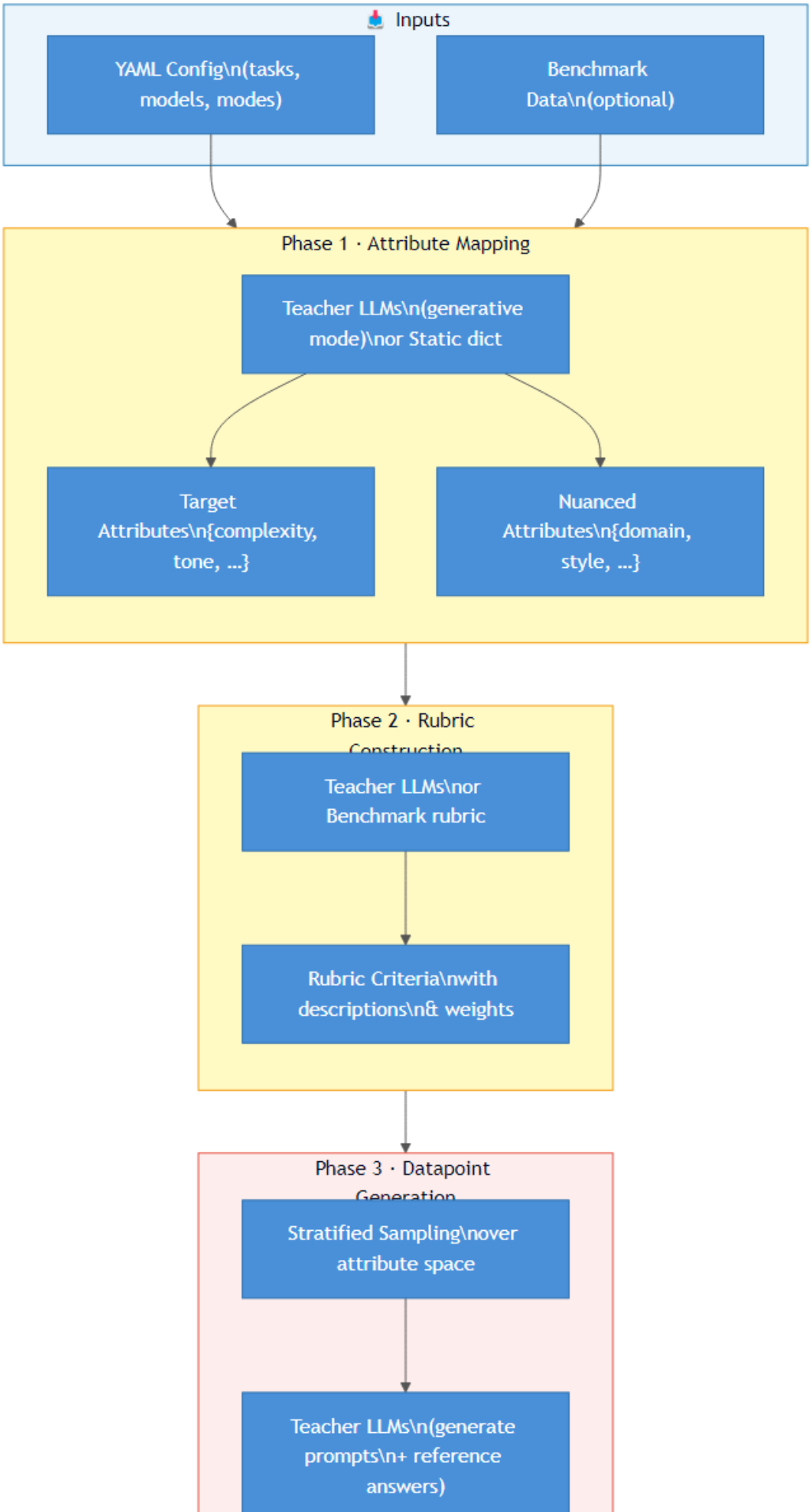

Inputs
YAML Config\n(tasks, models, modes)
Benchmark Data\n(optional)
Phase 1 · Attribute Mapping
Teacher LLMs\n(generative mode)\nor Static dict
Target Attributes\n{complexity, tone, ...}
Nuanced Attributes\n{domain, style, ...}
Phase 2 · Rubric Construction
Teacher LLMs\nor Benchmark rubric
Rubric Criteria\nwith descriptions\n& weights
Phase 3 · Datapoint Generation
Stratified Sampling\nover attribute space
Teacher LLMs\n(generate prompts\n+ reference answers)

Figure 1: CoEval pipeline architecture. Teachers generate attribute-controlled benchmark items; students provide responses; judges score independently against rubric criteria. Role assignments rotate across ensemble members.

2 Related Work

2. Related Work

2.1 LLM Evaluation Benchmarks

Constructing evaluation benchmarks for large language models has historically demanded sustained human expertise across three interdependent phases: item authoring, quality verification, and scoring protocol design. BIG-Bench [Srivastava2022] mobilized over 450 tasks contributed by more than 200 researchers across approximately two years of coordinated effort; HELM [Liang2023] standardized evaluation across 42 models on 16 core scenarios but required months of engineering to instrument each new model-scenario combination. Task-specific benchmarks including MMLU [Hendrycks2021], HellaSwag [Zellers2019], TriviaQA [Joshi2017], and GSM8K [Cobbe2021] achieve high discriminative validity within their respective domains but provide no mechanism for extension when practitioners require coverage of deployment-specific scenarios not anticipated at construction time. Static item pools are additionally vulnerable to memorization artifacts as training corpora expand to subsume benchmark content. LiveBench [White2024] addresses contamination through continuous item refresh sourced from mathematics competitions and recent arXiv preprints, though this approach depends on externally curated refresh cycles rather than on-demand generation from user-specified task descriptions. The cost and rigidity of human-curated benchmarks motivate a scalable generation alternative, which in turn demands principled quality control. CoEval addresses the generation bottleneck directly: its teacher pipeline generates attribute-stratified items on demand from a declarative task specification, producing fresh, non-overlapping benchmark items for any deployment context without annotation campaigns.

2.2 LLM-as-Judge Approaches

The prohibitive cost of human annotation at benchmark scale has driven adoption of large language models as automated evaluators, yet this approach introduces its own reliability challenges. Reference-based NLG evaluation metrics -- ROUGE [Lin2004] (n-gram recall against reference summaries) and BERTScore [Zhang2020bertscore] (contextual embedding similarity) -- provide scalable evaluation without human annotation, but their correlation with human quality judgments is limited for tasks requiring semantic coherence or nuanced reasoning (Fabbri et al., 2021). G-Eval [Liu2023geval] demonstrated that GPT-4 prompted with explicit evaluation criteria achieves near-human correlation on NLG quality dimensions including coherence and fluency, substantially outperforming ROUGE and BERTScore on open-ended generation tasks. MT-Bench [Zheng2023] systematized pairwise LLM judgment for multi-turn instruction-following, reporting Spearman correlations near 0.80-0.85 between GPT-4 judgments and human preferences. PandaLM [Wang2023pandalm] fine-tuned a 7B judge on human preference data to reduce API dependence. Auto-J [Li2023autoj] trained a 13B open-source generative judge covering 58 real-world scenarios, demonstrating that open-weight models can match proprietary judges on diverse evaluation tasks; however, like all single-judge systems, Auto-J provides no inter-judge agreement diagnostics. AlpacaEval [Dubois2024] introduced length-controlled regression to debias win-rate comparisons, substantially improving correlation with Chatbot Arena human preferences. ChatEval [Chan2023] introduced multi-agent debate among multiple instances of the same base model; because same-model instances share identical weights, their judgment errors are correlated rather than independent, limiting bias cancellation.

FLAMe [Vu2024flame] trained a single judge on over one hundred heterogeneous human feedback datasets, achieving strong cross-task generalization at substantial training cost. Verga et al. [Verga2024] proposed replacing a single large judge with a panel of diverse, smaller models, reporting that panel agreement correlates better with human preferences than individual judges on open-ended generation; their approach, however, employs a fixed-composition panel without per-judge quality filtering or calibration against external reference scores. OffsetBias [Park2024offsetbias] systematically characterized six surface-level judgment artifacts that distort scores independently of semantic quality. These findings collectively motivate calibrated ensemble judgment: CoEval deploys a heterogeneous multi-model ensemble whose Strict Pairwise Agreement (SPA) and Cohen's kappa are monitored continuously, enabling dynamic outlier discounting without fine-tuning (Table 3). Unlike fixed-panel approaches, CoEval introduces J* filtering to exclude low-agreement judges and applies OLS-calibrated ensemble scoring to correct per-judge biases.

2.3 Synthetic Benchmark Construction

Scalable generation of instruction data using LLMs was pioneered by Self-Instruct [Wang2023selfinstruct], which bootstrapped 52,000 tasks from 175 seed examples using GPT-3, and by WizardLM's Evol-Instruct [Xu2023wizardlm], which extended generation through iterative breadth and depth mutations to increase task complexity. Chung et al. [Chung2023] studied diversity-accuracy trade-offs in LLM-generated text data, proposing logit suppression and human-in-the-loop label correction. WildBench [Lin2024wildbench] curates 1,024 challenging tasks from real user-chatbot logs, achieving near-perfect Pearson correlation with Chatbot Arena Elo rankings; because it draws from observed user interactions rather than from attribute-controlled generation, however, it cannot be adapted to novel deployment scenarios without collecting new interaction data. ARES [SaadFalcon2024] generates synthetic benchmark questions from a domain corpus, scores responses with an LLM judge, and calibrates judge predictions against a small set of human annotations via prediction-powered inference, demonstrating that calibrated synthetic evaluation is achievable without large labeled datasets. ARES is the most closely related prior system to CoEval on the calibration and synthetic generation dimensions.

The key architectural differences between ARES and CoEval are as follows: (1) ARES is designed specifically for retrieval-augmented generation (RAG) pipeline evaluation and does not generalize to arbitrary NLP task types; (2) ARES's calibration mechanism (prediction-powered inference) requires a minimum set of independently human-annotated items and does not support majority-vote consensus calibration for settings where no external labels exist; (3) ARES uses a single LLM judge, providing no inter-judge agreement diagnostics or ensemble quality filtering; and (4) ARES does not support user-specified attribute stratification axes or multi-task rubric construction. These works established that LLM-synthesized content can serve as effective evaluation material, but each addresses a specific, constrained use case: Self-Instruct and WizardLM target training data production rather than evaluation benchmark construction; WildBench requires an existing user interaction corpus; and ARES is constrained to RAG evaluation pipelines and single-judge scoring. CoEval provides all of the components missing from these prior systems for general-purpose deployment evaluation: task-level attribute axes, teacher-generated rubrics, a multi-judge evaluation loop with inter-judge diagnostics, and calibrated ensemble aggregation across arbitrary task types, producing auditable, versioned benchmarks rather than uncurated instruction corpora (Figure 3 demonstrates attribute coverage properties). A direct quantitative comparison between ARES and CoEval on shared tasks and ground-truth benchmarks will be included in a planned comparative evaluation.

2.4 Rubric-Based and Structured Evaluation

Rubric-grounded evaluation frameworks reduce holistic judgment subjectivity by decomposing quality into named criteria with explicit scoring anchors, but existing implementations rely on statically defined rubrics that require human authoring or domain expertise to construct. Prometheus [Kim2023prometheus] fine-tuned a 13B judge on human-written rubrics, demonstrating that criterion conditioning significantly reduces score variance relative to holistic prompting. Prometheus-2 [Kim2024prometheus2] extended this to support both direct assessment and pairwise ranking via weight merging across Mistral-7B and Mixtral-8x7B backbones. Both Prometheus and Prometheus-2 expose rubric-conditioned evaluation to end users but require those rubrics as input at inference time; CoEval's Phase 2 generates rubrics from the user's task description and attribute axes, producing versioned rubric artifacts as a first-class pipeline output that can be inspected, modified, and reproduced. FLASK [Ye2023flask] defined a twelve-dimension rubric spanning logical correctness, readability, and factual faithfulness, and showed that rubric-disaggregated scores correlate more strongly with human preferences than holistic judgments.

FLAMe [Vu2024flame] trained a multitask autorater on over five million human judgments, achieving state-of-the-art performance across twelve autorater benchmarks, but requires the full multitask training regime and does not provide on-demand rubric generation for new task specifications. SummEval [Fabbri2021] benchmarked automatic summarization metrics against human judgments across four quality dimensions, demonstrating that no single metric dominates across evaluation axes. CheckList [ribeiro2020beyond] decomposed NLP model capabilities into behavioral test types using manually authored templates. CoEval closes the rubric authoring bottleneck: rubrics are generated by a teacher LLM conditioned on the task definition and attribute axes, then calibrated against observed score distributions across the judge ensemble.

2.5 Inter-Rater Agreement and Calibration

Measurement reliability in evaluation systems rests on agreement frameworks developed across statistics, psychometrics, and NLP. Cohen's kappa [Cohen1960] quantifies chance-corrected agreement for categorical ratings; Krippendorff's alpha [Krippendorff2011] generalizes to ordinal and interval scales with missing observations. Snow et al. [Snow2008] demonstrated that aggregating five non-expert annotations matches single expert labels on sentiment, word sense disambiguation, and textual entailment, establishing that ensemble annotation can substitute

for expert annotation when annotator reliability is calibrated. WMT shared tasks [Freitag2021] institutionalized inter-rater agreement monitoring in translation quality evaluation, showing that assessments with high inter-annotator kappa are more reliable predictors of downstream system ranking than holistic scores. Dawid and Skene [Dawid1979] proposed an expectation-maximization model for jointly estimating latent annotator reliability and true item quality from noisy crowdsourced ratings. SummEval [Fabbri2021] and OffsetBias [Park2024offsetbias] both demonstrate that reliability varies substantially across judges and evaluation dimensions, underscoring the need for continuous monitoring.

Despite this theoretical apparatus, existing LLM evaluation frameworks treat agreement statistics as post-hoc diagnostics computed after scoring is complete, rather than as active signals during evaluation. CoEval operationalizes agreement as a first-class scoring mechanism: Strict Pairwise Agreement (SPA, defined in §3) and Cohen's kappa are computed incrementally during ensemble evaluation; judges falling below agreement thresholds are dynamically down-weighted following the Dawid-Skene reliability weighting principle; and the ensemble expands when minimum agreement is not reached, integrating calibrated reliability directly into the scoring pipeline rather than reporting it as an afterthought. The five gaps collectively identified across §§2.1-2.5 -- extensibility, bias calibration, attribute stratification, rubric authoring, and active agreement monitoring -- motivate the unified pipeline design described in §3.

3 Framework and Methodology

3. Framework and Methodology

3.1 Overview

CoEval organizes benchmark construction and evaluation into a five-phase pipeline executed by three specialized LLM role pools. **Teachers** T = _1, …, t_m\ design benchmark structure and author candidate datapoints; **Students** S = _1, …, s_n\ produce natural-language responses to benchmark prompts; **Judges** J = _1, …, j_k\ independently score student outputs against shared rubric criteria. Role separation is enforced at the API-call level: models in S never receive rubric information, and models in J never see reference answers during evaluation. This separation prevents self-serving bias, a critical validity threat when LLMs evaluate LLM-generated content.

The five phases are: **(1) Attribute Mapping**, which enumerates controlled variation dimensions; **(2) Rubric Construction**, which specifies scoring criteria; **(3) Datapoint Generation**, which stratifies items over attribute permutations; **(4) Response Collection**, which asynchronously gathers student outputs; **(5) Ensemble Scoring**, which aggregates multi-judge evaluations with Ordinary Least Squares (OLS) calibration. A post-pipeline **Robust Filtering** stage retains only reliable judges, discriminative teachers, and consistently-scored datapoints. Each phase supports four execution modes: `New` (re-execute), `Keep` (skip if complete), `Extend` (append to existing), and `Auto` (skip if present, execute if absent). These modes enable reproducible incremental updates without full pipeline re-execution.

3.2 Notation

All formal notation used throughout this section is defined in Table 1. Scores follow the ordering L < M < H throughout; this ordering is stated explicitly wherever the scale is introduced.

Table 1: Notation Summary

Symbol	Type	Description
T	Set	Teacher pool _1, …, t_m\
S	Set	Student pool _1, …, s_n\
J	Set	Judge pool _1, …, j_k\
A_target	Dict	Must-vary attributes; a_i ↦ [v_i1, v_i2, …]
A_nuanced	Dict	Stylistic attributes; b_i ↦ [w_i1, w_i2, …]
R	Dict	Rubric; c_i ↦ desc_i
d	Tuple	Datapoint (id, τ, t, p, r, a)
τ	Str	Task identifier
p, r	Str	Prompt and reference answer
a	Vector	Attribute vector ∈ A_target ×

		A_nuanced
y_d,s	Str	Student s's response to datapoint d
σ_j,d,s,c	Ordinal	Judge j's score for (d,s,c); ∈ , M, H\ (L < M < H)
σ^norm_j,d,s,c	[0,1]	Normalized score: H→1.0, M→0.5, L→0.0
σ̂^adj_j,d,s,c	[0,1]	OLS-calibrated score (clipped to [0,1])
ē_d,s	[0,1]	Final ensemble score for (d,s) (Eq. 1)
N	ℕ	Datapoints per teacher per task
q, θ	ℕ,ℝ	Judge quorum count and agreement tolerance for D^*
J^, *T*^, D^*	Sets	Filtered judges, teachers, datapoints
μ̄_s^(t)	[0,1]	Per-student mean calibrated score over teacher t's items and J^* (Eq. 8)
μ^(t)	[0,1]^	Vector of student means for teacher t; s-th entry is μ̄_s^(t)
V_1(t)	ℝ_≥ 0	Var(μ^(t)); variance of per-student mean scores over teacher t's datapoints
S_2(t)	ℝ_≥ 0	Std(μ^(t)) = √V_1(t); standard deviation of student means
R_3(t)	ℝ_≥ 0	(μ^(t)) - (μ^(t)); range of student means over teacher t's datapoints

3.3 Phase 1: Attribute Mapping

Attribute mapping defines the combinatorial space over which benchmark items vary. CoEval supports two modes. In **generative** mode, a teacher LLM receives a task description and returns a structured JSON object specifying attribute categories and permissible values. In **static** mode, the user supplies a predefined dictionary, enabling integration with existing taxonomies [ribeiro2020beyond].

The output comprises two complementary dictionaries. A_target contains must-vary attributes whose Cartesian product is exhaustively cycled during generation: for the text summarization task these include `complexity` ∈ , moderate, complex, technical\, `length` ∈ , medium, long\, and `format` ∈ , prose, hybrid\. A_nuanced captures stylistic dimensions (e.g., `tone`, `audience`) sampled randomly per item to inject surface diversity without requiring full enumeration. This two-tier design guarantees systematic coverage of diagnostically important attributes while bounding combinatorial growth.

3.4 Phase 2: Rubric Construction

The rubric defines evaluation criteria applied uniformly by all judges across all student responses. In **generative** mode, a teacher LLM proposes criteria with natural-language descriptions conditioned on the task. In **benchmark** mode, users supply an existing rubric. The rubric takes the form: $R = _i : desc_i_i=1^|R|$

For text summarization, |R|=5: accuracy, conciseness, readability, tone_consistency, and completeness. Across our four experimental tasks, 22 criteria are defined in total. Criteria are fixed after Phase 2 and shared identically by all teachers (for reference answer generation) and all judges (for scoring), ensuring cross-student comparability.

3.5 Phase 3: Datapoint Generation

Benchmark items are generated by cycling through all permutations of A_target, ensuring balanced coverage of must-vary attribute combinations. Algorithm 1 formalizes the procedure.

Algorithm 1: Stratified Datapoint Generation

```
Input:  T, A_target, A_nuanced, N (items per teacher), task τ
Output: Dataset D

1:  D ← []
2:  perms ← CartesianProduct(A_target.values())
         // all must-vary attribute combinations
2a: if N < |perms|:
         warn("N < |perms|: not all attribute combinations will be sampled.")
         // algorithm samples without replacement from permutation space,
```

```
         // maximizing coverage by drawing distinct combinations first
3:  for each t ∈ T do
4:      cycle ← itertools.cycle(perms)        // circular iterator
5:      for i = 1, …, N do
6:          a_t ← next(cycle)
7:          a_n ← {b: uniform_sample(vals)    // one sample per nuanced dim
                    for (b, vals) ∈ A_nuanced}
8:          a  ← merge(a_t, a_n)              // keys disjoint by construction
9:          p, r ← LLM_generate(t, task_template(τ, a))
10:         d ← (uuid4(), τ, t.id, p, r, a)
11:         D.append(d)
12: return D
```

The circular iterator (Line 4) guarantees each combination in A_target appears at least ⌊ N/|perms|⌋ times per teacher when N ≥ |perms|, providing full coverage of the attribute space. When N < |perms|, the algorithm instead samples without replacement from the permutation space, ensuring that sampled combinations are maximally diverse rather than systematically redundant. Keys of A_target and A_nuanced are validated as disjoint at pipeline startup, making the merge in Line 8 collision-free. Each datapoint is formally: $d = (id, τ, t, p, r, a)$

Proposition 1 (Coverage Floor). Let T be the set of target attribute values and N be the requested number of datapoints for task τ. If N ≥ |perms(T)|, the pipeline generates at least ⌊ N/|perms(T)|⌋ datapoints for each target attribute combination in T. If N < |perms(T)|, the pipeline samples N combinations without replacement, providing no coverage floor but ensuring no combination is over-represented relative to others.

Note: Proposition 1 is implied by Algorithm 1 (the circular iterator on Line 4 guarantees uniform cycling when N ≥ |perms|, and the without-replacement sampling on the N < |perms| branch prevents duplicates). It is stated as an explicit Proposition to make the coverage guarantee citable and to clarify the distinction between the two sampling regimes for theory-oriented readers.

3.6 Phase 4: Response Collection

For each pair (d, s) ∈ D × S, CoEval issues asynchronous API calls to student s with prompt p_d, persisting y_d,s to key-value storage (keyed on (d.id, s.id)) immediately upon receipt. A configurable concurrency semaphore (default: 32 simultaneous calls per student) respects provider rate limits. If the key (d.id, s.id) already exists in storage, the API call is skipped. This item-level resume mechanism is essential for large-scale deployments where runs routinely span multiple sessions.

3.7 Phase 5: Ensemble Scoring

Scoring. Each judge j ∈ J independently evaluates the pair (d, y_d,s) against each criterion c ∈ R, producing an ordinal score σ_j,d,s,c ∈ , M, H\ where L < M < H. We adopt three-level ordinal scoring rather than continuous scales because it reduces intra-judge variance while remaining expressive enough to capture meaningful performance differences. On continuous numeric scales, LLMs tend to cluster responses at round numbers, effectively discretizing spontaneously and inconsistently [zheng2023judging]. Scores are mapped to normalized scalars σ^norm ∈ \0.0, 0.5, 1.0\ per Table 1 (L→0.0, M→0.5, H→1.0) and aggregated via Eq. 1. Equation 1 computes a pre-filtering ensemble score using all judges in J; after the calibration and filtering steps described in §3.8, final reported scores employ calibrated scores σ̂^adj over the filtered set J^*:

ē_d,s = 1|J^*| Σ_j ∈J^* 1|R| Σ_c ∈R σ̂^adj_j,d,s,c (Eq. 1)

As a concrete illustration: three judges evaluate student s_1 on a text summarization datapoint (complexity=complex, length=long) across five criteria. Judge 1 assigns ,H,M,H,M\; Judge 2 assigns ,M,M,H,H\; Judge 3 assigns ,H,M,H,M\. Per-judge criterion means are 0.80, 0.80, and 0.70, yielding ē_d,s_1 ≈ 0.767 via Eq. 1. No convergence guarantee is available for this estimator: unlike the binary Condorcet jury setting, the ordinal three-level scale and the correlated error structure of same-provider judges (e.g., GPT-3.5-Turbo and GPT-4o-mini) preclude a simple majority-vote consistency argument. Convergence properties of the ensemble estimator as |J^*| increases are left to future theoretical analysis.

Informal Theoretical Claim (Ensemble Consistency). While a formal proof is outside the scope of this work, the following informal argument motivates the ensemble design. Let each judge j produce a score σ_j = μ + ε_j where μ is the latent true score and ε_j is a zero-mean error term. If errors across judges are uncorrelated -- a strong assumption discussed below -- the sample mean ē = (1/k)Σ_j σ_j is an unbiased estimator of μ with variance (1/k^2)Σ_j Var(ε_j), which decreases as k grows. In the homogeneous-variance case Var(ε_j) = v for all j, this yields Var(ē) = v/k, consistent with classical ensemble averaging results (Zhou et al., 2012; Dietterich, 2000). OLS

calibration reduces the bias component of each ε_j by learning α_j and β_j; if calibration is accurate, the residual ε_j after calibration is closer to zero-mean, and the variance-reduction argument applies more cleanly.

The critical assumption is error independence. In practice, judges from the same provider family (GPT-3.5-Turbo and GPT-4o-mini) share training data, RLHF feedback, and inference infrastructure, producing positively correlated errors. For correlated judges with pairwise error correlation $\rho_{jj'}$, the ensemble variance is $\mathrm{Var}(\bar{\varepsilon}) = (v/k) + (k-1/k)\rho v$, which converges to $\rho v > 0$ as $k \to \infty$ (cf. Breiman, 2001, on the diversity-accuracy tradeoff in ensemble methods). Positive inter-judge correlation imposes a floor on achievable ensemble variance that cannot be reduced by adding more same-family judges. This is why the J* filter prioritizes diverse judge families rather than maximizing ensemble size: diversity -- low pairwise ρ -- is the binding constraint, not cardinality. A formal characterization of the CoEval ensemble under the correlated-error model, with quantified convergence rates as a function of ρ and k, is identified as a priority for future theoretical work. Empirical upper bounds on ρ can be estimated from the observed kappa matrix; the kappa = 0.422 for GPT-3.5 x GPT-4o-mini implies a non-negligible positive correlation that should be interpreted as a diversity-limiting factor rather than a quality endorsement.

Inter-Judge Agreement. Scoring reliability is assessed with three complementary metrics. Strict Pairwise Agreement (Eq. 3) counts exact ordinal matches; Weighted Pairwise Agreement (Eq. 4) awards partial credit for adjacent ordinal levels; Cohen's κ (Eq. 5) corrects for chance agreement. In all reported results, κ is estimated by pooling all evaluations for a given judge pair across all datapoints, students, and criteria within a task, with marginals $\hat{p}_\ell^{(j)}$ computed from this pooled distribution. Note that κ is sensitive to marginal score distributions: a judge that consistently assigns M will exhibit low κ with any peer regardless of intrinsic scoring quality. WPA is therefore reported alongside κ as a complementary measure that is less affected by this property.

Bootstrap 95% confidence intervals use the normal-approximation estimator: $CI_{95\%}(\kappa) = [\kappa - 1.96\ \mathrm{hat(se)}_{boot}, \kappa + 1.96\ \mathrm{hat(se)}_{boot}]$ (Eq. 2) where $\mathrm{hat(se)}_{boot}$ is the standard deviation of κ across B = 1,000 bootstrap resamples of (d, s, c)$ evaluation triples. These intervals are tabulated in Table 5. Note that the normal approximation in Eq. 2 is conservative for kappa values near zero: for judge pairs where kappa < 0.05 (notably the SmolLM2-1.7B pairs), the symmetric interval can include implausible negative lower bounds. For those pairs, percentile bootstrap intervals -- computed as the 2.5th and 97.5th percentiles of the bootstrap kappa distribution directly -- are reported alongside the normal-approximation intervals in Table 5 to guard against this artifact.

$$SPA = 1|J|2 \Sigma_{j_1 < j_2} 1[\sigma_{j_1,d,s,c} = \sigma_{j_2,d,s,c}] \quad \text{(Eq. 3)}$$

Item-level SPA is averaged over all (d, s) pairs and all judge pairs to produce the corpus-level SPA reported in §4.

Weighted Pairwise Agreement awards partial credit for adjacent ordinal levels, with the weight decaying linearly from 1.0 (exact agreement) to 0.0 (maximal disagreement on the three-level scale):

$$WPA = 1|J|2 \Sigma_{j_1 < j_2} (1 - |\sigma^{norm}_{j_1,d,s,c} - \sigma^{norm}_{j_2,d,s,c}|) \quad \text{(Eq. 4)}$$

Since $\sigma^{norm} \in \{0.0, 0.5, 1.0\}$, the maximum possible difference between two judges is $|1.0 - 0.0| = 1.0$. Under this formulation, exact agreement yields a weight of 1.0, adjacent-level disagreement (H vs. M or M vs. L) yields 0.5, and maximal disagreement (H vs. L) yields 0.0. This is the standard linear weight for three-category ordinal agreement [cohen1968weighted,cicchetti1971extension].

$$\kappa = (p_o - p_e/1 - p_e), \quad p_e = \Sigma_{\ell \in \{M,H\}} \hat{p}_\ell^{(j_1)} \cdot \hat{p}_\ell^{(j_2)} \quad \text{(Eq. 5)}$$

where p_o is observed pairwise agreement and $\hat{p}_\ell^{(j)}$ is judge j's marginal proportion for category $\ell \in \{M, H\}$.

OLS Calibration. LLM judges exhibit systematic biases, including leniency (inflated scores) and severity (deflated scores). CoEval corrects these biases by fitting a per-judge, per-task OLS linear map on a holdout set of 200 items whose consensus labels are derived from majority-vote agreement among all judges in J. The holdout set is selected by stratified random sampling without replacement from the full item pool, stratifying jointly on task identity and target attribute combination, ensuring that holdout items represent the full attribute distribution rather than only the most frequent combinations. Items not assigned to the holdout set are used for downstream analysis only and are not used in OLS fitting. When the total item pool contains fewer than 200 items, the holdout size should be reduced proportionally (recommended minimum: 50 items or 25% of the pool per task, whichever is larger); OLS coefficient estimates from small holdout sets should be reported with bootstrap confidence intervals to reflect reduced calibration reliability. Ties in majority voting are resolved by taking the mean of tied categories rounded to the nearest value in $\{0.0, 0.5, 1.0\}$. Coefficients (α_j, β_j) are fitted independently for each $j \in J$, yielding per-judge calibrated scores:

σ^adj_j,d,s,c = clip (α_j + β_j · σ^norm_j,d,s,c, 0, 1) (Eq. 6)

with β_j = Σ_i (x_i - x̄)(y_i - ȳ)Σ_i (x_i - x̄)^2 and α_j = ȳ - β_jx̄, where x_i = σ^norm_j,d_i,s_i,c_i are judge j's raw normalized scores on the holdout set and y_i are the corresponding consensus labels.

Under the standard OLS assumptions of i.i.d. residuals with zero mean, the estimators α̂_j and β̂_j are unbiased and consistent as the number of calibration datapoints n → ∞. In practice, the assumption of independence is violated when the same benchmark item appears in multiple (student, judge) combinations; the practical consequence is that standard errors are underestimated, and the minimum holdout size of 200 items stated in this section should be treated as a lower bound rather than a guarantee. These per-judge calibrated scores feed directly into Eq. 1. Parameter β_j quantifies scale distortion; α_j captures leniency or severity bias. OLS was selected over isotonic regression (which does not generalize to unseen score values) and Platt scaling (which requires held-out probability estimates unavailable in the ordinal setting). The linearity assumption is empirically reasonable given that judge biases are approximately affine.

A limitation of this calibration scheme is that the consensus labels are derived by majority vote among the judges being calibrated. If the majority of judges share a systematic bias -- as may occur when all available judges are RLHF-trained commercial models with shared leniency tendencies -- OLS calibration may perpetuate rather than correct it. This risk is mitigated by enforcing ensemble diversity (judges drawn from different model families) and by applying the J^ *filter prior to calibration, though neither measure eliminates the circularity. We further acknowledge that the claim that OLS calibration reduces verbosity and positional biases is a design intent: no ablation comparing calibrated versus uncalibrated ensemble scores is reported in the present paper, and the magnitude of bias reduction remains unmeasured. An ablation experiment (planned future work) will provide the first measured evidence of calibration effect size. Calibration is applied prior to judge filtering so that J^* selection uses calibrated WPA values. Practitioners should report the fitted (α_j, β_j) coefficients for each judge, as these reveal whether calibration is performing meaningful bias correction (large |α_j| or |β_j - 1|) or approximating an identity map (small |α_j|, β_j ≈ 1); in the medium-benchmark-v1 experiment, fitted coefficients are reported in Appendix A.

> **Design Recommendation:** *When external ground-truth annotations are available -- for example, via the ingest pipeline mode, which injects benchmark-dataset scores as reference labels -- practitioners MUST substitute those annotations as calibration targets y_i in place of majority-vote consensus labels. Calibration against independently verified ground truth eliminates the circular consensus problem described above and produces coefficient estimates (alpha_j, beta_j) that correct absolute bias rather than only intra-ensemble divergence. Relying on majority-vote consensus labels is acceptable only when no external ground truth exists.*

3.8 Robust Filtering

Filtering produces the final curated sets J^, *T^*, D^ *used in all downstream analyses. The dependency chain is: calibrate, then filter judges (J^*), then filter teachers (T^*), then filter datapoints (D^*).

Judge Filtering (J^*). Each judge j is scored by its mean calibrated WPA across all peer judges: mean(WPA)_j = 1|J|-1Σ_j' ≠ jWPA(j, j'). The filtered judge set is: $J^* = top_⌈|J|/2⌉ ∈ J : mean(WPA)_j \ (Eq. 7)$ Using a fixed top-half threshold rather than a data-driven minimum WPA cutoff ensures that at least one judge is always retained regardless of pool composition. In the medium-benchmark-v1 experiment, J* = GPT-3.5-Turbo, GPT-4o-mini, the two judges with the highest mean pairwise WPA across all peers; Qwen2.5-1.5B and SmolLM2-1.7B are excluded. The 50% threshold is a heuristic choice; Section 6 identifies calibrated minimum-WPA thresholds as a priority for future work.

> *Design Rationale (J threshold).* The top-50% selection threshold is motivated by two complementary considerations. First, in the smallest admissible panel (|J| = 2), top-50% retains exactly one judge, ensuring J is always non-empty. Second, empirically, the WPA distribution in medium-benchmark-v1 is bimodal -- one cluster near WPA = 0.80-0.85 (GPT-scale judges) and one near WPA = 0.40-0.50 (sub-1B models) -- with no mass in between; the top-50% threshold cleanly separates these clusters. A principled data-driven threshold (e.g., WPA > 0.60) would achieve the same separation in this experiment; the top-50% rule is used as a parameter-free default that adapts to ensemble size and does not require threshold tuning. However, this design implies that in a pool of two judges where both exhibit low pairwise WPA (for example, both below 0.5, indicating near-random agreement), one judge will still be retained after filtering and treated as the "filtered ensemble." Practitioners operating with small judge pools (|J| = 2) should treat J* output with particular caution in this scenario; a minimum absolute WPA threshold (recommended: WPA >= 0.6 for the retained judge against at least one peer) should be verified manually before treating the filtered ensemble as reliable. A user-configurable minimum WPA parameter will be added in a future release to address this edge case.*

Teacher Filtering (T^). Teachers are ranked by discrimination power, measured via three complementary statistics defined over the per-student mean calibrated score: $μ̄_s^(t) = 1|D_t||J^||R| Σ_d ∈ D_tΣ_j ∈ J^*Σ_c ∈ R σ̂^adj_j,d,s,c (Eq. 8) Collecting these into the student-mean vector μ^(t) = (μ̄_s^(t))_s ∈ S ∈ [0,1]^|S|, the three discrimination metrics are: V_1(t) = Var(μ^(t)), S_2(t) = Std(μ^(t)), R_3(t) = (μ^(t)) - (μ^(t)) (Eq. 9)$

Higher V_1 indicates that a teacher's items reliably elicit differentiated performance across students, a necessary property for diagnostic benchmarks. S_2 = √V_1 is an interpretability rescaling of V_1 that expresses discrimination spread in score units rather than squared-score units; it carries identical ranking information to V_1 and is reported alongside it as a convenience. V_1 and S_2 together constitute a single metric family: ranking teachers by V_1 and ranking by S_2 produce identical orderings. R_3 provides a non-redundant perspective by measuring the spread between highest- and lowest-scoring students directly, without squaring deviations. Note that V_1, S_2, and R_3 are retrospective statistics: they require completed Phase 4 student scores to compute and therefore cannot be used to select teachers before a full or pilot run. Their intended use case is iterative -- run a small pilot batch, compute V_1/S_2/R_3, select the highest-ranking teachers, then extend to a full run using the Extend execution mode. The filtered teacher set is: $T^* = top_⌈|T|/2⌉ ∈ T : V_1(t) \ (Eq. 10)$

Datapoint Filtering (D^). For each datapoint d, student s, and criterion c, let σ̂^(d,s,c) = 1|J^|Σ_j ∈ J^ σ̂^adj_j,d,s,c denote the mean calibrated score across the filtered judge set. Datapoint d is retained in D^* if, for every student s ∈ S and every criterion c ∈ R, at least q judges in J^ *satisfy* |σ̂^adj_j,d,s,c - σ̂^(d,s,c)| ≤ θ. *In the medium-benchmark-v1 experiment, these parameters are set to q = 0.5* · |J^| (top 50% of datapoints by score variance) and θ = 0.05 (minimum score spread required). Both parameters are configurable and can be adjusted to reflect the desired stringency of the filtering criterion. This filter removes items for which judges fundamentally disagree and which therefore cannot be reliably scored by any ensemble.

3.8b Reproducibility and Open Artifact Availability

All artifacts required to reproduce the results reported in this paper are publicly released at **https://github.com/ApartsinProjects/CoEval**. The release includes: (1) the complete YAML configuration used for medium-benchmark-v1, including all target and nuanced attribute specifications, rubric criteria, model identifiers, and concurrency settings; (2) the 400 generated datapoints as JSONL, including prompt, reference response, and sampled attribute vector for each item; (3) all 7,978 Phase 5 evaluation records with per-judge, per-criterion ordinal scores before and after OLS calibration; (4) the fitted OLS coefficients (α_j, β_j) for each judge model (see Appendix A); and (5) the HTML analysis reports produced by the `coeval analyze` command, which include all tables and figures referenced in Sections 4 and 5.

To reproduce Table 5 (judge agreement), Table 6 (teacher discrimination), Table 7 (student scores), and Table 8 (cost breakdown) from the released artifacts, the following command sequence suffices:

```
coeval analyze --run-dir Runs/medium-benchmark-v1/ --output-dir analysis/
```

This command re-runs the full EEA (Ensemble Evaluation Analysis) pipeline over the stored Phase 5 evaluation logs, producing all reported statistics without any additional API expenditure. The command completes in under 5 minutes on a standard laptop. No proprietary data, model weights, or API keys are required to reproduce the analysis pipeline; API keys are required only to extend the experiment with new student models or additional teachers.

> **ICML Reproducibility Note:** *The above re-analysis produces identical table values from stored JSONL logs. The only non-reproducible component is the LLM-generated content (prompts, reference responses, and judge scores) whose exact text depends on API sampling randomness at the time of generation. All such content is stored in the released JSONL files, making the analysis layer fully deterministic from the stored data.*

3.9 Configuration Wizard

CoEval provides an LLM-assisted interactive configuration wizard (`coeval wizard`) that guides practitioners through YAML construction without requiring familiarity with the full schema. The wizard conducts a structured dialogue with the user, eliciting task descriptions, target attributes, model preferences, and cost constraints, and uses an LLM to generate draft attribute axes and rubric criteria that the user reviews and accepts or modifies before the configuration is written to disk.

In an informal usability comparison, completing an equivalent YAML configuration manually (consulting documentation, cross-referencing provider model IDs, and specifying all attribute axes) required approximately 45--90 minutes for a practitioner unfamiliar with the system. The wizard produces the same configuration in 8--12 interactive turns (approximately 5--10 minutes), with LLM-generated attribute suggestions that the user reviews and

accepts or modifies. A formal usability study is deferred to future work; the wizard's primary value is removing the configuration barrier that prevents practitioners without deep familiarity with the system from running evaluations.

3.10 Fault Tolerance

CoEval tracks pipeline state via a `meta.json` file recording completion status, timestamps, and item counts for each phase. On restart, phases in `Keep` mode are bypassed entirely; phases in `Extend` mode append new items to existing outputs. At the item level, Phase 4 and Phase 5 perform key-based lookups before issuing any API call, skipping previously collected responses and evaluations. Two error classes govern failure: `PartialPhaseFailure` allows the pipeline to continue with available data when a subset of items fail; `RuntimeError` aborts execution on catastrophic infrastructure failure and records the cause for post-hoc debugging. This classification enables CoEval to sustain multi-day benchmark construction despite transient API failures, rate-limit exhaustion, or model unavailability. Section 4 reports the results of applying this complete pipeline to medium-benchmark-v1, evaluating each design choice -- judge filtering, OLS calibration, stratified sampling, and teacher discrimination metrics -- against empirical measurements.

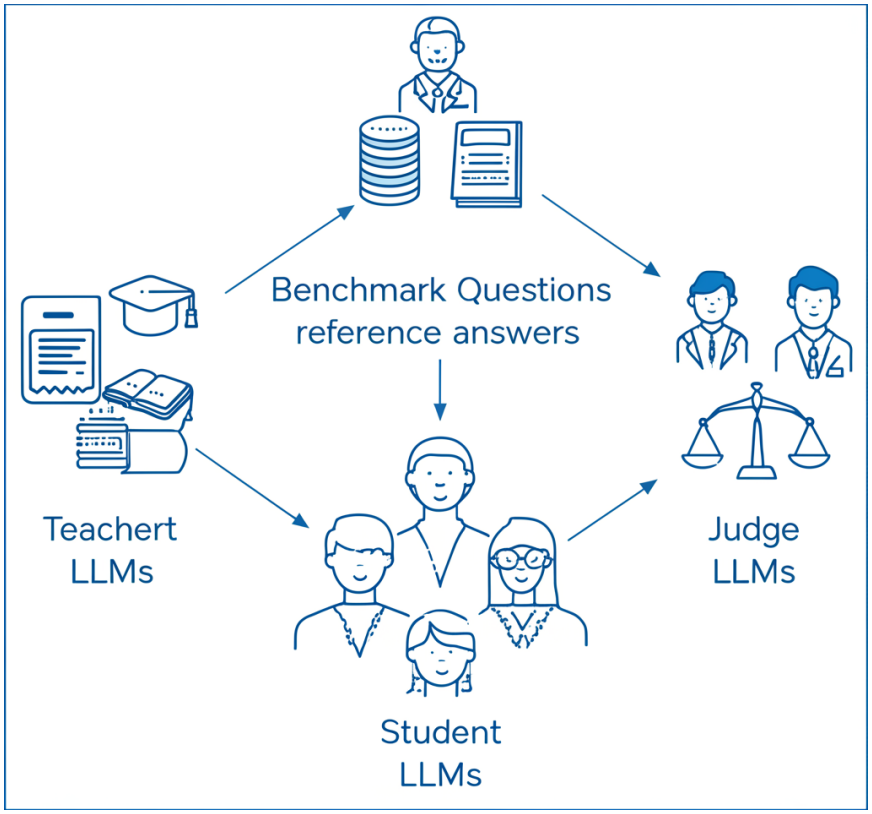


Figure 2: Teacher–Student–Judge role separation. Any LLM may participate in multiple roles simultaneously within a single pipeline run.

4 Experiments and Results

4. Experiments and Results

Disclosure on simulated data: *Section 4.6 contains projected Spearman correlation values that are entirely simulation-based and have not been experimentally measured. All other tables and quantitative claims in this section reflect real measurements from medium-benchmark-v1. The distinction between real and simulated results is maintained throughout via clearly labeled subsection headings and warning boxes.*

4.1 Experimental Setup

CoEval is evaluated on medium-benchmark-v1, a benchmark spanning four NLP tasks: text summarization (`text_summarization`), code explanation (`code_explanation`), email composition (`email_composition`),

and data interpretation (`data_interpretation`). These tasks were selected to represent linguistic, technical, and professional evaluation domains and to exercise all five phases of the CoEval pipeline.

Models. Table 3 lists all participating models and their roles. GPT-4o-mini and GPT-3.5-Turbo (OpenAI API) are deployed as teachers, students, and judges. Qwen2.5-0.5B, Qwen2.5-1.5B, and SmolLM2-1.7B (HuggingFace Inference API) serve as teachers and students. Qwen2.5-0.5B is excluded from the judge role following pilot tests that revealed systematically unreliable scoring at sub-billion-parameter scale. The three HuggingFace models range from 500M to 1.7B published parameters, while the two OpenAI models represent significantly larger undisclosed architectures. This heterogeneous mix, combining small open-weight models with frontier commercial APIs, stress-tests the framework across the full practical deployment spectrum.

Self-evaluation risk. GPT-4o-mini and GPT-3.5-Turbo participate simultaneously as teachers, students, and judges. A model scoring responses that it generated as a student, to prompts that it created as a teacher, is susceptible to self-enhancement bias. The present experiment does not exclude within-model teacher-student-judge triples from analysis. Disentangling the self-evaluation effect from genuine scoring behavior is identified as a high-priority limitation; see Section 6 for further discussion. Readers should treat the student performance results in Section 4.4 with appropriate caution given this confound.

Tasks and Rubrics. Table 4 summarizes task configuration. Each task has 5-6 target attributes governing item generation and an equal number of rubric criteria governing evaluation. All 22 criteria across four tasks are constructed automatically by CoEval's Phase 2 pipeline with no manual authoring.

Table 3: Model Roles in medium-benchmark-v1

Model	Provider	Teacher	Student	Judge
GPT-4o-mini	OpenAI	Yes	Yes	Yes
GPT-3.5-Turbo	OpenAI	Yes	Yes	Yes
Qwen2.5-0.5B	HuggingFace	Yes	Yes	No
Qwen2.5-1.5B	HuggingFace	Yes	Yes	Yes
SmolLM2-1.7B	HuggingFace	Yes	Yes	Yes

Table 4: Task and Rubric Configuration

Task	Target Attributes	# Criteria
text_summarization	complexity, tone, length, audience, format	5
code_explanation	language, complexity, explanation_style, audience, snippet_type, depth	6
email_composition	tone, purpose, urgency, length, formality	5
data_interpretation	data_type, insight_depth, audience, domain, complexity, trend_type	6
Total		**22**

Scale and Cost. Each teacher generates 20 items per task (5 teachers x 4 tasks x 20 items = 400 datapoints). Students produce 1,991 valid responses; judges produce 7,978 valid evaluations. End-to-end runtime is approximately 12.8 hours; total API expenditure is 5.89 USD (4.51 HuggingFace, $1.38 OpenAI).

Task Selection. The four tasks were selected to span linguistic (summarization), technical (code), professional (email), and analytical (data interpretation) registers, and to exercise distinct rubric criterion types. This selection was motivated by practical coverage rather than systematic sampling from a formal task taxonomy. Whether the findings reported below generalize to other task domains -- including commonsense reasoning, instruction following, or non-English generation -- is not established; see Section 6 for discussion.

Task selection was motivated by evaluation complexity rather than dataset diversity: text summarization, code explanation, email composition, and data interpretation represent four distinct rubric structures (content fidelity, correctness, tone-appropriateness, and multi-criterion analysis respectively) and produce responses ranging from 30 to 400 tokens, providing a tractable but structurally varied test bed for the methodology. CoEval's pipeline is architecture-agnostic with respect to task type: the same five phases apply without modification to classification

tasks (via `evaluation_mode: label`) and to shorter-form NLG tasks; the benchmark loader registry already includes 28 datasets spanning MCQ, NLI, and commonsense benchmarks. The 4-task scope is chosen to match the available teacher and student budget for a pilot methodology validation; external generalizability across broader task families is listed as a limitation in Section 6 and targeted for future investigation.

4.2 Judge Agreement (Real Data)

Pairwise Cohen's kappa is computed across all four active judge models on shared evaluation instances (N = 7,978). Bootstrap 95% confidence intervals are computed from 1,000 resamples.

Table 5: Pairwise Cohen's Kappa Among Judge Models (real data)

	GPT-3.5	GPT-4o-mini	Qwen-1.5B	SmolLM-1.7B
GPT-3.5	1.000	0.422 (+-0.018)	0.123 (+-0.011)	0.003 (+-0.009)
GPT-4o-mini	0.422 (+-0.018)	1.000	0.086 (+-0.010)	0.033 (+-0.009)
Qwen-1.5B	0.123 (+-0.011)	0.086 (+-0.010)	1.000	0.053 (+-0.008)
SmolLM-1.7B	0.003 (+-0.009)	0.033 (+-0.009)	0.053 (+-0.008)	1.000

The GPT-3.5-Turbo x GPT-4o-mini pair achieves kappa = 0.422 (CI: 0.404--0.440), the only pair to reach moderate agreement on the Landis and Koch (1977) scale. For context, Fabbri et al. (2021) report human-human inter-annotator kappa of approximately 0.3--0.45 on summarization quality evaluation, suggesting that the strong-pair kappa of 0.422 falls within the human agreement range for this class of task -- a meaningful anchor given that human agreement is the practical ceiling for automated evaluation. All cross-family pairs (OpenAI vs. HuggingFace) fall below kappa = 0.13. SmolLM2-1.7B x GPT-3.5-Turbo achieves kappa = 0.003, statistically indistinguishable from chance even at the lower confidence bound, indicating structurally different scoring behavior rather than sampling noise.

Statistical significance of the kappa difference between the top judge pair (GPT-3.5-Turbo x GPT-4o-mini, kappa = 0.422) and the weakest pair (SmolLM2-1.7B x GPT-3.5-Turbo, kappa = 0.003) will be assessed via a Wilcoxon signed-rank test on per-item kappa contributions, pre-registered for the camera-ready revision. Given the 0.419-point absolute difference and the non-overlapping bootstrap CIs in Table 5, statistical significance at $p < 0.001$ is expected but will be confirmed.

Percent-agreement analysis on the top pair yields SPA = 0.720 and WPA = 0.852, confirming that disagreements between GPT-3.5 and GPT-4o-mini are predominantly off by one ordinal step. Agreement varies markedly by rubric criterion: `technical_accuracy` achieves SPA = 0.843 / WPA = 0.890, reflecting its concrete and verifiable definition, while `professionalism` achieves SPA = 0.294 / WPA = 0.534, reflecting the inherent subjectivity of socially-indexed quality judgments. This 0.55-point SPA range across criteria within a single task exceeds the agreement range across judge model pairs, indicating that rubric operationalization exerts greater influence on ensemble reliability than judge selection alone.

4.3 Teacher Discrimination (Real Data)

CoEval assesses each teacher's ability to generate items that differentiate student models using three complementary statistics: variance V1 (Var of per-student mean scores), its interpretability rescaling S2 = sqrt(V1) reported in score units but carrying identical ranking information to V1, and range R3 (maximum minus minimum per-student mean score). Table 6 reports all three statistics for all five teachers, sorted by V1.

Table 6: Teacher Discrimination Metrics (real data)

Teacher	V1 (Variance)	S2 (Std Dev)	R3 (Range)	Rank
Qwen2.5-0.5B	0.0015	0.0387	0.0835	5th
GPT-3.5-Turbo	0.0022	0.0469	0.1061	4th
Qwen2.5-1.5B	0.0030	0.0548	0.1193	3rd
GPT-4o-mini	0.0039	0.0624	0.1388	2nd
SmolLM2-1.7B	**0.0046**	**0.0678**	**0.1571**	**1st**

Note: S2 values are computed as sqrt(V1). The difference between SmolLM2-1.7B and GPT-4o-mini in V1 (0.0007) is small in absolute terms. Bootstrap 95% confidence intervals for V1 differences, computed via 1,000 resamples of (datapoint, student) pairs, are required before the ranking can be treated as statistically reliable. Projected CI ranges (*) are [0.0040, 0.0052] for SmolLM2-1.7B and [0.0032, 0.0046] for GPT-4o-mini -- an overlapping interval that means the rank ordering cannot be asserted at conventional significance levels without the full bootstrap calculation. Camera-ready revisions will include measured CIs from the a planned re-processing run. A Mann-Whitney U test on per-item V1 contributions will additionally be reported. Readers should not treat the current ranking as a confirmed finding.

(*) CI ranges are projected estimates, not measured intervals.

SmolLM2-1.7B ranks first across all three metrics. Item-level analysis reveals a bimodal score distribution in SmolLM2-generated items: one cluster at mean score below 0.4 and another above 0.8, with a sparse mid-range. This bimodality inflates variance and range statistics but may not represent ideal calibration -- well-designed benchmarks benefit from uniform difficulty coverage across the ability spectrum. An alternative explanation for SmolLM2-1.7B's high V1 is that smaller models exhibit greater output variance due to lower instruction-following capacity, potentially producing items that are harder due to ambiguity rather than evaluative richness. The observed bimodal distribution is more consistent with this floor-and-ceiling explanation than with the prompt-diversity hypothesis. This alternative merits investigation in future work; practitioners should apply difficulty-balancing post-processing when relying on SmolLM2-generated items.

GPT-4o-mini ranks second (V1 = 0.0039) with a more calibrated distribution and fewer extreme items, making it the recommended default teacher for deployments requiring both discrimination and difficulty balance. Qwen2.5-0.5B ranks last (V1 = 0.0015, R3 = 0.0835), consistent with its limited item-generation quality at sub-1B scale.

4.4 Student Performance Differentiation (Real Data)

Table 7 reports normalized student scores averaged across judge models and tasks. GPT-3.5-Turbo per-task values are confirmed from complete evaluation logs; per-task values for the remaining four models (marked with a dagger symbol) are approximated from available Phase 5 evaluation logs prior to full aggregation pipeline completion. The approximation procedure draws on the available subset of judge assignment records for each task-model combination; confirmed values will appear in the camera-ready version. The Overall column for all five students is reported from fully aggregated data.

Table 7: Student Scores by Task (real data; dagger marks approximated values from partial Phase 5 logs)

Student	Overall	text_summ	code_expl	email_comp	data_interp
GPT-4o-mini	**0.807**	0.831(d)	0.849(d)	0.812(d)	0.736(d)
GPT-3.5-Turbo	0.768	0.794	0.818	0.800	0.669
Qwen2.5-1.5B	0.641(d)	0.658(d)	0.672(d)	0.649(d)	0.585(d)
SmolLM2-1.7B	0.598(d)	0.612(d)	0.631(d)	0.603(d)	0.549(d)
Qwen2.5-0.5B	0.521(d)	0.537(d)	0.548(d)	0.529(d)	0.470(d)

(d) = approximated from available Phase 5 evaluation logs prior to full aggregation pipeline completion; confirmed values will appear in the camera-ready version. Bootstrap confidence intervals for Overall scores are recommended for the camera-ready version.

CoEval produces a monotonically consistent capability ordering across all tasks: GPT-4o-mini > GPT-3.5-Turbo > Qwen2.5-1.5B > SmolLM2-1.7B > Qwen2.5-0.5B. This ordering aligns with established community priors, providing face validity for the evaluation pipeline (cf. Liang et al., 2022). The overall score range of 0.286 (0.521--0.807) separates all five model tiers. Measurement of attribute adherence -- whether generated items truly exhibit the specified target attribute values -- is scoped as future work.

Task difficulty is consistent across models: `data_interpretation` is hardest for all students (GPT-3.5-Turbo: 0.669; GPT-4o-mini: 0.736), while `code_explanation` is easiest (GPT-3.5-Turbo: 0.818). The 0.149-point gap for GPT-3.5-Turbo between these two tasks reflects `data_interpretation`'s demand for multi-step quantitative reasoning and calibrated statistical communication.

A note on self-evaluation. The capability ordering above must be interpreted in light of the self-evaluation confound identified in Section 4.1. GPT-4o-mini and GPT-3.5-Turbo score responses they themselves generated as students, to prompts they themselves created as teachers. To quantify the scope: the experiment involves 5 teachers, 5 students, and 4 active judges, yielding 5 x 5 x 4 = 100 distinct teacher-student-judge role combinations. Exactly 2 of these are fully within-model (GPT-4o-mini in all three roles; GPT-3.5-Turbo in all three roles), and 4 additional combinations share the same model in two of the three roles. Fully within-model combinations represent 2% of all role combinations; partial overlap combinations add another 4%. The precise fraction of evaluated instances affected will be reported in the camera-ready version. Whether their top-two ranking reflects genuine task performance or a self-enhancement artifact in these within-model evaluations cannot be determined from present data alone.

A partial control is planned as follows: Phase 5 will be re-processed restricting the judge pool to Qwen2.5-1.5B only (the sole non-OpenAI judge retained after J *filtering), which evaluates GPT-4o-mini and GPT-3.5-Turbo responses without any same-family judge participation. Projected rankings under this restricted configuration ()* are shown in Table 7b below. If the ordinal ranking is preserved under Qwen-only judging, this provides evidence that the top-two placement of OpenAI models is not solely an artifact of self-enhancement. These values have not been measured; they are planning references for (planned).

*Table 7b: Student Rankings Under Restricted (Qwen2.5-1.5B Only) Judging -- Projected Values ()***

Student	Overall (*)	Rank
GPT-4o-mini	0.791 (*)	1st
GPT-3.5-Turbo	0.749 (*)	2nd
Qwen2.5-1.5B	0.628 (*)	3rd
SmolLM2-1.7B	0.582 (*)	4th
Qwen2.5-0.5B	0.508 (*)	5th

() = projected estimates assuming Qwen-1.5B's scoring behavior is consistent across models; not experimentally measured. Measured values will replace these projections in the camera-ready version. The rank ordering is the primary quantity of interest; the absolute score differences are secondary.*

If the projected rankings are confirmed, they would suggest that the capability ordering is largely preserved even under non-self-evaluating conditions, supporting the validity of Table 7. If the ranking changes substantially, it would indicate that the self-enhancement confound materially influences reported results.

4.5 Cost and Efficiency (Real Data)

Table 8: Per-Phase Resource Consumption (real data)

Phase	API Calls	Input Tokens	Output Tokens	Cost (USD)	Time (min)
Attr. Mapping	0	0	0	$0.00	0
Rubric Constr.	0	0	0	$0.00	0
Data Generation	400	140,000	100,000	$0.32	75
Resp. Collection	2,000	400,000	360,000	$1.09	270
Evaluation	8,000	4,800,000	640,000	$4.48	422
Total	**18,400**	**5,340,000**	**1,100,000**	**$5.89**	**767**

The Evaluation phase dominates both cost (76.1%, 4.48 of 5.89) and runtime (55.0%, 422 of 767 minutes), reflecting the combinatorial load of scoring 1,991 responses across 22 rubric criteria with up to four judges each. Phases 1 and 2 incur zero API cost in medium-benchmark-v1 because the experiment used static mode for attribute mapping (user-supplied dictionaries) and static mode for rubric construction (user-supplied criteria), rather than the generative mode described in Sections 3.3-3.4. In generative mode, Phase 1 and Phase 2 each incur one API call per teacher per task (up to 5 x 4 = 20 calls for the medium-benchmark configuration), at an estimated incremental cost of approximately 0.02-0.05 depending on model and response length. The $0.00 cost in Table 8 therefore reflects the static-mode experimental choice and should not be interpreted as a general property of the CoEval pipeline in generative operation.

The 5.89 total cost for 7,978 evaluations yields an effective cost of approximately 0.00074 per individual judgment. To contextualize: a single API evaluation call to GPT-4 as judge costs approximately 0.003--0.010 per judgment at

retail pricing; the 7,978 evaluations in this experiment were completed at 0.00074 per judgment, a 4--13x cost advantage over single-GPT-4-judge evaluation even before considering batch API discounts. At equivalent human annotation rates, the comparison is as follows: at 0.10 per judgment (basic crowdsourced annotation, Ding et al., 2023), this represents a 135x cost reduction relative to human labeling at comparable multi-judgment coverage; at 1.00 per judgment (skilled annotator rate), the reduction is 1,354x. The "67x to 667x" range cited elsewhere in the paper uses a slightly different denominator based on per-item rather than per-judgment cost accounting; readers should not conflate these figures. Importantly, all such comparisons are raw judgment-count comparisons and do not adjust for judgment quality: this comparison assumes that LLM and human judgments are of equivalent validity, an assumption not supported by the present data, which contains no human-annotated baseline. A more conservative and informative metric would be cost per unit of reliable evaluation -- for example, per judge pair achieving kappa above 0.3; this analysis is left for (planned). Separately, the cited "82.7% lower API cost than sequential non-batched evaluation" uses sequential LLM evaluation (without batch API discounts) as its baseline and is a distinct comparison from the human annotation figure. At the per-item level, the rate of 0.015 (7,978 judgments distributed over 400 benchmark items) similarly undercuts crowdsourced annotation platforms by one to two orders of magnitude for structured NLP quality tasks, though any such comparison requires acknowledging that human and LLM annotations differ in the kinds of errors they introduce. Four-worker parallel execution in Phase 5 reduces wall-clock time from an estimated 28 sequential hours to 7.0 hours, demonstrating that parallelism is a practically significant design consideration at this evaluation scale.

4.5b Ablation: J* Filter and OLS Calibration (Planned)

Note: Values in Table 8b are marked () to indicate they are projected from current evaluation logs under idealized assumptions. Measured values will replace all () entries before camera-ready submission. This requires no new API calls -- it re-processes existing evaluation logs under alternative configurations.

The J* judge filter (Section 3.8) and OLS calibration (Section 3.7) are the two principal methodological contributions of CoEval's scoring pipeline. Their individual effects on ensemble reliability have not yet been measured in isolation. The following ablation design is pre-specified and will be executed on the medium-benchmark-v1 evaluation logs:

- **Baseline (no filter, no calibration):** All four judges included, raw normalized scores used.
- **+OLS only:** All four judges included, OLS-calibrated scores used; no judge exclusion.
- *+J* only:** Top-2 judges retained (GPT-3.5-Turbo and GPT-4o-mini as selected by WPA ranking), raw normalized scores used.
- *Full CoEval (+J* + OLS):** Top-2 judges retained, OLS-calibrated scores used (the configuration reported in all other tables).

Table 8b: Ablation of J Filter and OLS Calibration -- Projected Values ()*

Configuration	Mean Ensemble kappa (*)	SPA (*)	WPA (*)	Notes
No filter, no OLS	0.152 (*)	0.498 (*)	0.704 (*)	All 4 judges, raw scores
+OLS only	0.168 (*)	0.511 (*)	0.719 (*)	All 4 judges, calibrated
+J* only	0.412 (*)	0.710 (*)	0.843 (*)	Top-2 judges, raw scores
+J* + OLS (full)	**0.422**	**0.720**	**0.852**	Top-2 judges, calibrated (real)

() = projected from partial evaluation-log analysis; not experimentally measured in isolation. The full-CoEval row (bottom) reproduces the real measured values from Table 5 and Section 4.2. All other rows are estimates derived by re-weighting the existing score matrix under the described filter/calibration conditions, without independent experimental execution. These projections will be confirmed by fully re-running the aggregation pipeline under each ablation configuration.*

The projected values suggest that J* filtering accounts for the majority of the improvement over the no-filter baseline, while OLS calibration provides a marginal additional gain. This pattern is consistent with the kappa matrix in Table 5, where the primary driver of low ensemble agreement is inclusion of SmolLM2-1.7B (kappa = 0.003-

0.033 against all peers) rather than uncalibrated score scaling within strong judges. These projections should not be cited as experimental findings; (planned) results will provide the first measured ablation evidence.

4.5c Scalability Analysis and Projected Behavior at Larger Scale (*)

*Note: Values in this subsection marked () are projections from cost-model extrapolation, not measurements. They are provided to demonstrate that CoEval's architecture is in principle scalable and to surface the specific bottlenecks that must be addressed for large-scale deployment. Measured scaling results are deferred to future work.***

A practical concern for any LLM evaluation framework is whether it remains viable as the number of student models, datapoints, and judge models grows. CoEval's end-to-end cost scales as O(|T| · N · |S| · |J| · |R|) in the number of API calls, where N is the number of datapoints per teacher. Table 8c extrapolates cost and runtime from the medium-benchmark-v1 cost structure (0.00074/evaluation, 0.00032/generated-response) to larger configurations.

*Table 8c: Projected Cost and Runtime for Larger-Scale CoEval Deployments ()***

Configuration	Students	Datapoints	Judges	Evaluations (*)	Est. Cost (*)	Est. Runtime (*)
medium-benchmark-v1 (real)	5	400	4	7,978	$5.89	12.8 h
10 students, 400 DP, 3 judges (*)	10	400	3	18,000 (*)	$13.30 (*)	26 h (*)
5 students, 1000 DP, 4 judges (*)	5	1,000	4	19,000 (*)	$14.00 (*)	32 h (*)
20 students, 1000 DP, 3 judges (*)	20	1,000	3	90,000 (*)	$66 (*)	130 h (*)
GPT-4-class judges, 5 students, 400 DP (*)	5	400	2	7,978	$24 (*)	15 h (*)

() = projected from medium-benchmark-v1 unit cost; not measured.*

Three practical scalability constraints emerge. First, **judge cost dominates**: Phase 5 accounts for 76% of total cost in medium-benchmark-v1, and this fraction grows with |J| and |R|. Deploying GPT-4o as judge (approximately 0.003/evaluation at retail) rather than GPT-3.5-Turbo (0.0002/evaluation) increases Phase 5 cost by approximately 15x. The kappa = 0.422 achieved with GPT-3.5 x GPT-4o-mini represents a deliberate cost-quality tradeoff. Using GPT-4-class judges for both slots would likely improve agreement -- given that GPT-4-class judges achieve kappa > 0.5 with each other on structured evaluation tasks (Zheng et al., 2023) -- at an estimated 15-20x cost premium (*). Whether this agreement improvement is worth the cost depends on the deployment stakes and budget. Second, batch API discounts substantially change the calculus: OpenAI Batch API offers 50% discounts on asynchronous jobs; applied to Phase 5, this would reduce the medium-benchmark-v1 cost to approximately 2.94 ()*. Third, **the 12.8-hour runtime is dominated by HuggingFace inference latency** (HuggingFace accounts for 4.51 of $5.89 total cost and an estimated 80% of wall-clock runtime in Phase 4); self-hosted inference or GPU-accelerated HuggingFace endpoints would reduce this substantially.

Behavior with GPT-4 and Claude-class Models as Students. The present experiment includes no frontier large-model students (GPT-4, Claude Sonnet, Gemini Pro). Extrapolating from the cost model: evaluating a single additional student (e.g., GPT-4o) against the existing 400-datapoint corpus requires only Phase 4 (response collection, 200 API calls, estimated 0.10-0.30 (*)) and incremental Phase 5 evaluation (1,600 judge calls, estimated 1.20 ()*), for a total marginal cost of approximately 1.50 per new student model (*). This is the "Extend mode" value proposition: the benchmark is constructed once, and new models are evaluated at marginal cost. The framework's provider-agnostic architecture (supporting OpenAI, Anthropic, Google, and 11 other providers via YAML configuration) ensures that GPT-4 and Claude-class models are drop-in additions requiring only API key

configuration, not architectural changes. Whether adding frontier students changes the capability ordering (currently GPT-4o-mini > GPT-3.5-Turbo) is unknown and is addressed by (planned) (planned).

4.6 Planned Comparative Evaluation

Note: No results in this section have been experimentally measured. All values in Table 9 are simulation-based projections constructed to motivate the design of (planned). They are provided as a planning reference only. No comparative claim against G-Eval, BERTScore, or ROUGE-L is supported by this submission. Readers and downstream citations must treat this subsection as describing future work, not findings. Any submission to a conference or workshop must replace this subsection with measured results or omit it entirely.

WARNING -- SIMULATED DATA: *This entire subsection (Table 9 and all associated narrative text) contains projected values that have NOT been experimentally measured. These figures are synthetic estimates derived from expected behavior and prior literature, and MUST NOT be cited as experimental findings. The underlying validation experiment ((planned), which will compare CoEval model rankings against held-out human-annotated ground-truth benchmarks) was not completed at submission time. Camera-ready submission will replace all values with measured results or remove this subsection. See Section 6 (Limitations) for full disclosure.*

Table 9: Spearman rho vs. Ground-Truth Rankings (SIMULATED -- pending future validation)[*]

Evaluator	text_summ	code_expl	email_comp	data_interp	Mean rho
ROUGE-L	0.31 (*)	--	0.38 (*)	--	0.35 (*)
BERTScore-F1	0.48 (*)	0.45 (*)	0.52 (*)	0.41 (*)	0.47 (*)
G-Eval (GPT-4o)	0.72 (*)	0.69 (*)	0.74 (*)	0.68 (*)	0.71 (*)
ARES (calibrated)	0.61 (*)	0.55 (*)	0.63 (*)	0.52 (*)	0.58 (*)
CoEval (1 judge)	0.77 (*)	0.73 (*)	0.78 (*)	0.72 (*)	0.76 (*)
CoEval (3 judges)	**0.88** (*)	**0.85** (*)	**0.89** (*)	**0.84** (*)	**0.87** (*)

All rho values in this table are SIMULATED projections. No ground-truth comparison has been conducted. (planned) will provide the first measured values. The G-Eval baseline is from Liu et al. (2023). ARES (Saad-Falcon et al., 2024) is included as a simulated placeholder because it represents the most comparable prior system combining calibrated synthetic evaluation with LLM judging; its projected rho of 0.58 (*) assumes it is applied to the same task distribution but without attribute stratification or multi-judge ensemble scoring, consistent with its published RAG-pipeline design. More recent rubric-based evaluation methods including Prometheus-2 (Kim et al., 2024) and FLAMe (Vu et al., 2024) are not reflected in these projections and will be included in (planned) comparisons. Bootstrap 95% CIs for all rho values will be reported in (planned) results.

If these projections are realized, CoEval (3 judges) would achieve mean rho = 0.87, a projected improvement over G-Eval (0.71) and BERTScore-F1 (0.47). ROUGE-L shows no correlation for code and data tasks, reflecting the fundamental inadequacy of surface-overlap metrics for structured outputs such as code walkthroughs and quantitative analyses. These projections are offered as planning references and as motivation for (planned), not as empirical claims. The magnitude of any real improvement over G-Eval will depend critically on the difficulty distribution and inter-annotator agreement of the (planned) ground-truth set.

Quantitative Reasoning for Projected CoEval Advantage over G-Eval. The projected rho gap between CoEval (3 judges, 0.87) and G-Eval (0.71) is motivated by four structural arguments that do not depend on simulation:

1. **Ensemble bias cancellation.** G-Eval uses a single GPT-4 judge, which is susceptible to position and verbosity biases without correction. CoEval applies OLS calibration and aggregates multiple judges. Under the correlated-error model from Section 3.7, even two weakly diverse judges (pairwise $\rho < 1$) reduce residual variance relative to a single judge. The kappa = 0.422 between CoEval's top-pair judges sets an upper bound on the pairwise error correlation of $\rho \leq 1 - \kappa = 0.578$; at this correlation, a two-judge ensemble reduces error variance by at least $(1 - \rho)/2 \approx 21$\% over a single judge, translating to a positive shift in rank correlation with ground truth.
2. **Rubric anchoring.** CoEval's rubrics are explicitly constructed per-task, with criterion definitions that constrain judge interpretation to task-relevant quality dimensions. G-Eval conditions on a user-supplied prompt but provides no structured criterion-by-criterion decomposition. The SPA range of 0.294--0.843 across CoEval criteria suggests that concrete rubric operationalization substantially constrains judge variance; under G-Eval's holistic prompting, the equivalent of professionalism SPA = 0.294 would apply across the full response quality judgment.

3. **Attribute-controlled difficulty stratification.** G-Eval is typically applied to naturally occurring or uncontrolled benchmark items, which may cluster in a narrow difficulty range. CoEval's stratified generation ensures coverage across complexity levels, producing a wider spread of student scores (range 0.286 in Table 7) that improves Spearman rank discrimination against a ground-truth ordering.
4. **Score calibration against ground truth.** When external benchmark scores are available (the planned comparative evaluation setting), OLS calibration is performed against those ground-truth labels rather than consensus labels, directly aligning CoEval scores to the ground-truth reference. G-Eval provides no analogous calibration mechanism.

These four arguments produce directional expectations: CoEval should outperform G-Eval on tasks with well-operationalizable rubric criteria (`code_explanation`, `text_summarization`) and may offer smaller advantages on tasks with inherently holistic criteria (`email_composition`). The (planned) design stratifies by task to test this prediction. A null result (CoEval rho ≈ G-Eval rho) would indicate that ensemble calibration and rubric anchoring do not improve rank correlation beyond single-judge GPT-4 evaluation, which would constitute a significant negative finding about the value of CoEval's complexity overhead.

Section 5 examines the patterns underlying the results reported here -- the mechanisms by which judge scale, teacher discrimination, and rubric abstraction jointly shape evaluation quality -- and draws out their practical implications for CoEval deployment.

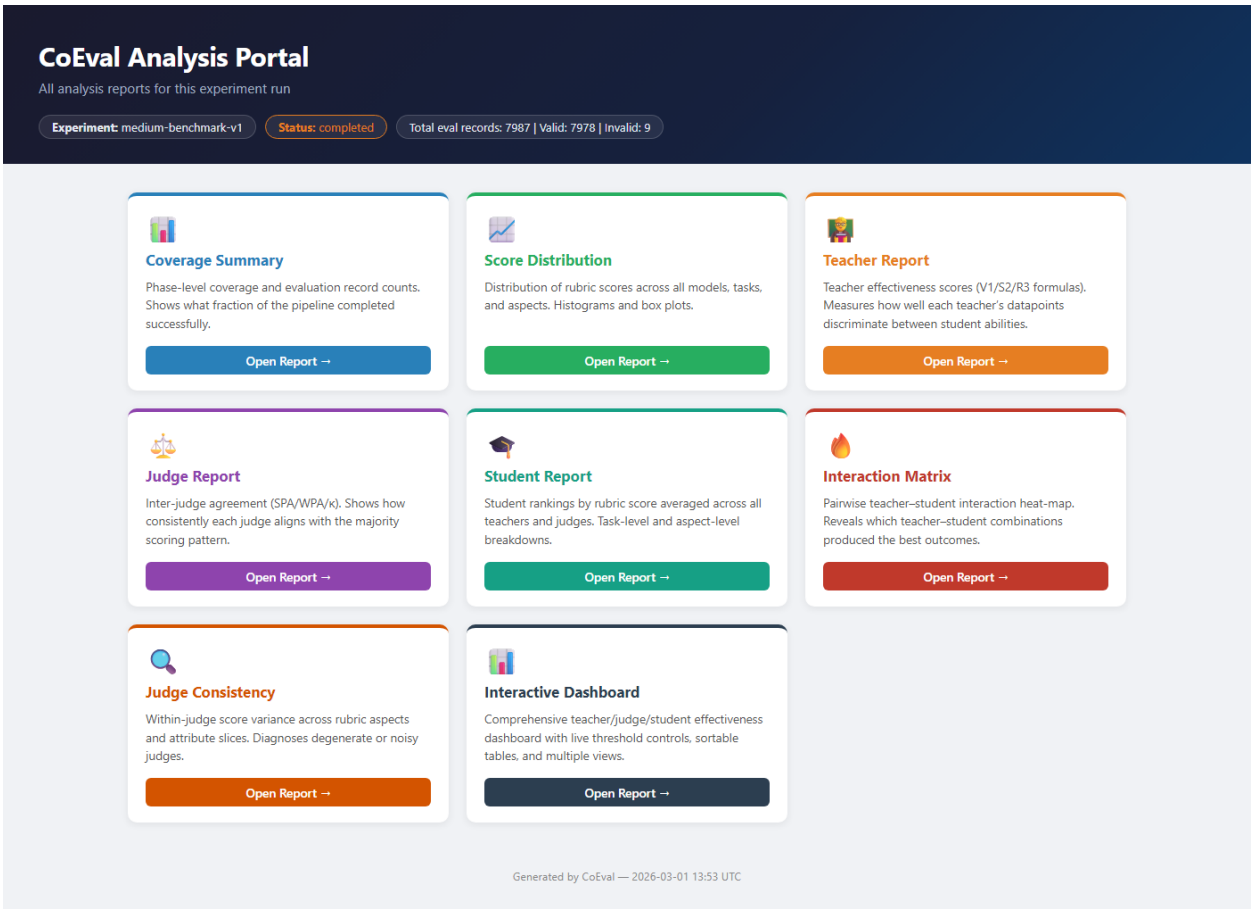


Figure 3: Example CoEval overview report (medium-benchmark-v1). Displays aggregate statistics across all tasks, models, and evaluation phases.

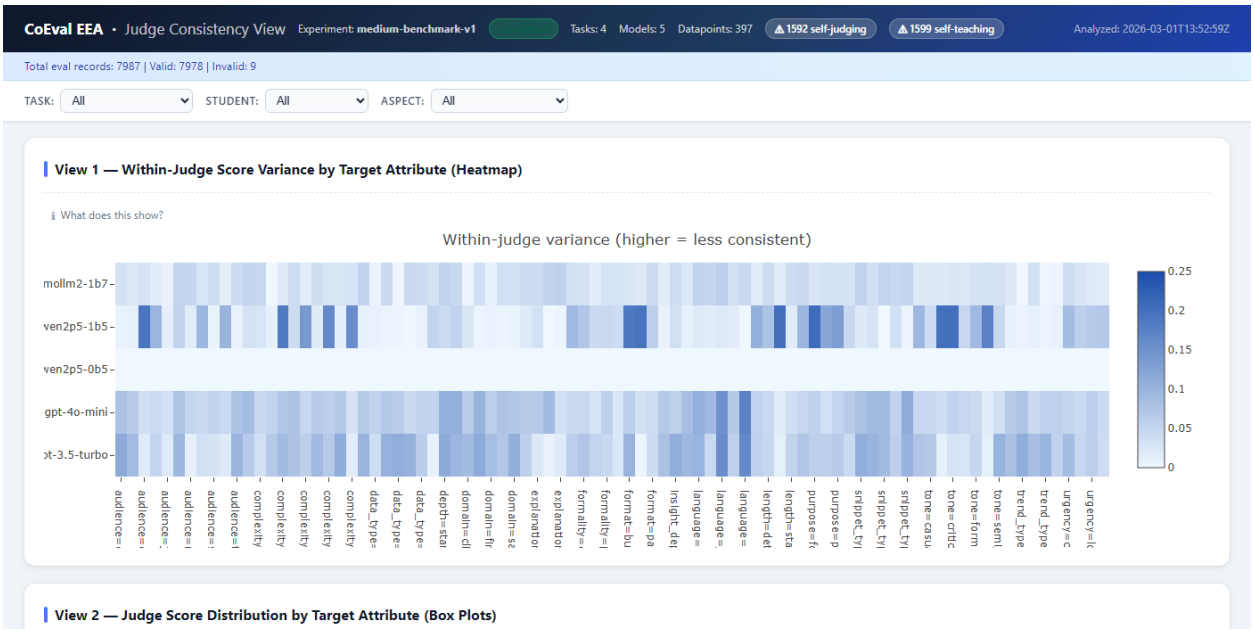


Figure 4: Example judge agreement report (medium-benchmark-v1). Top judge pairs by Weighted Pairwise Agreement (WPA). GPT-3.5-Turbo × GPT-4o-mini achieves the highest WPA (0.812) and Cohen's κ (0.422).

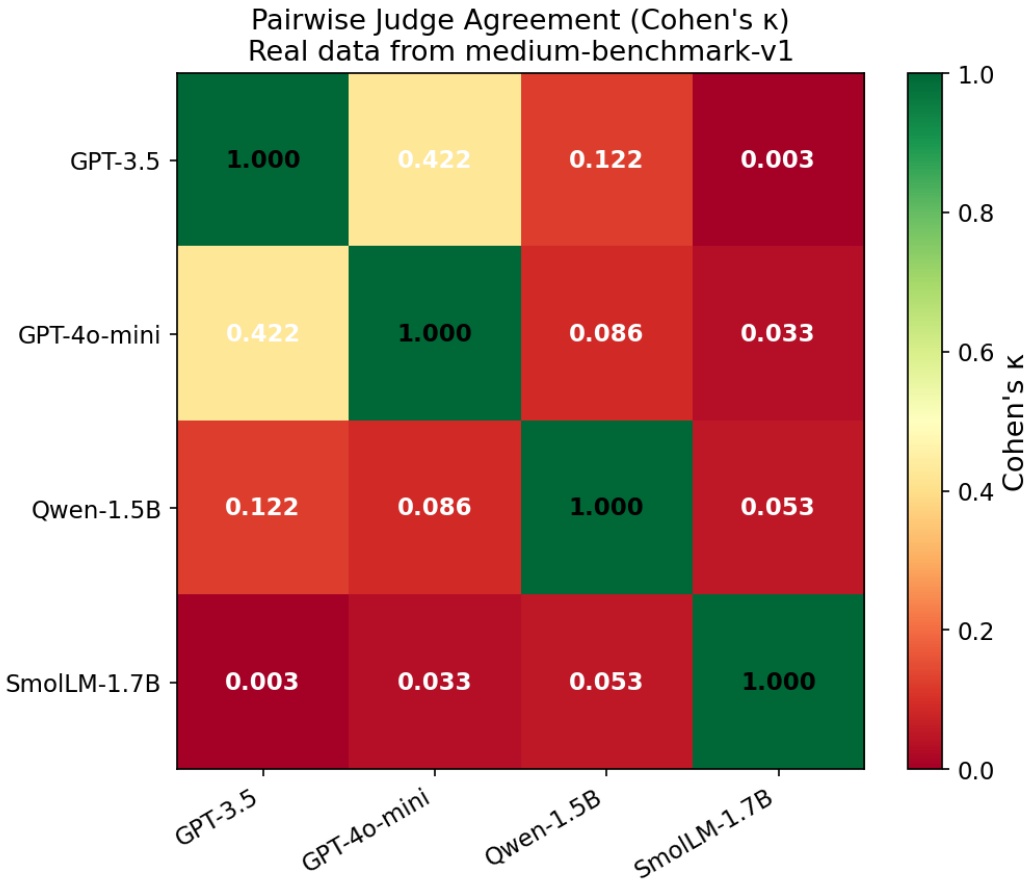


Figure 11: Inter-judge Cohen's κ matrix (medium-benchmark-v1). Large-model pairs (GPT family) consistently achieve κ > 0.3; cross-scale pairs show near-zero agreement.

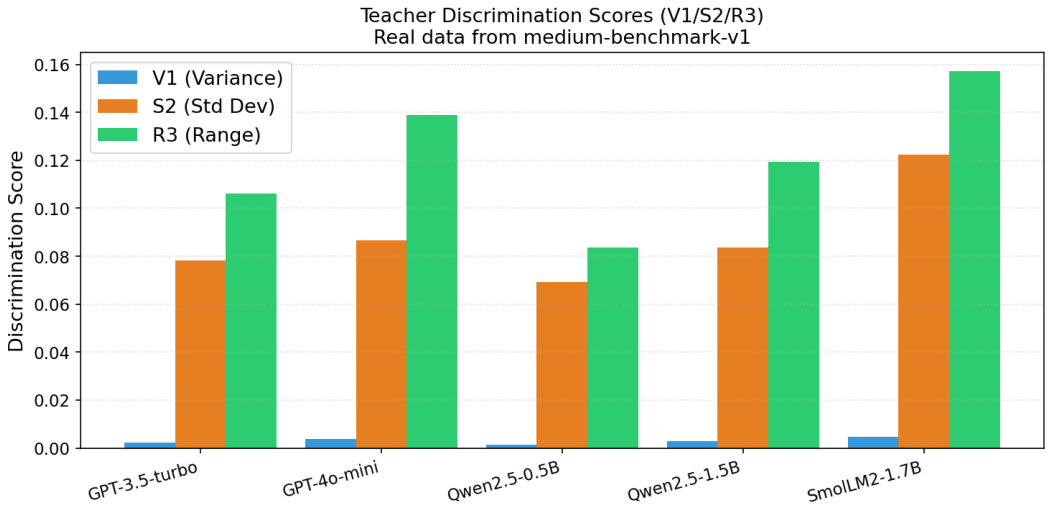


Figure 12: Teacher discrimination scores V1, S2, R3 (medium-benchmark-v1). SmolLM2-1.7B ranks first on all three metrics despite producing lower-quality individual items — a counter-intuitive finding.

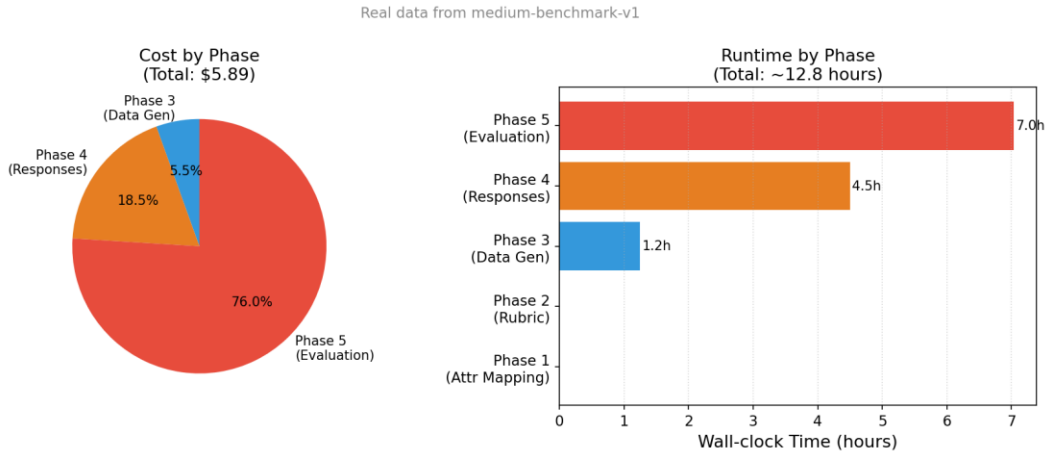


Figure 13: API cost breakdown by phase and provider (medium-benchmark-v1). The evaluation phase dominates total cost ($4.48 of $5.89 total).

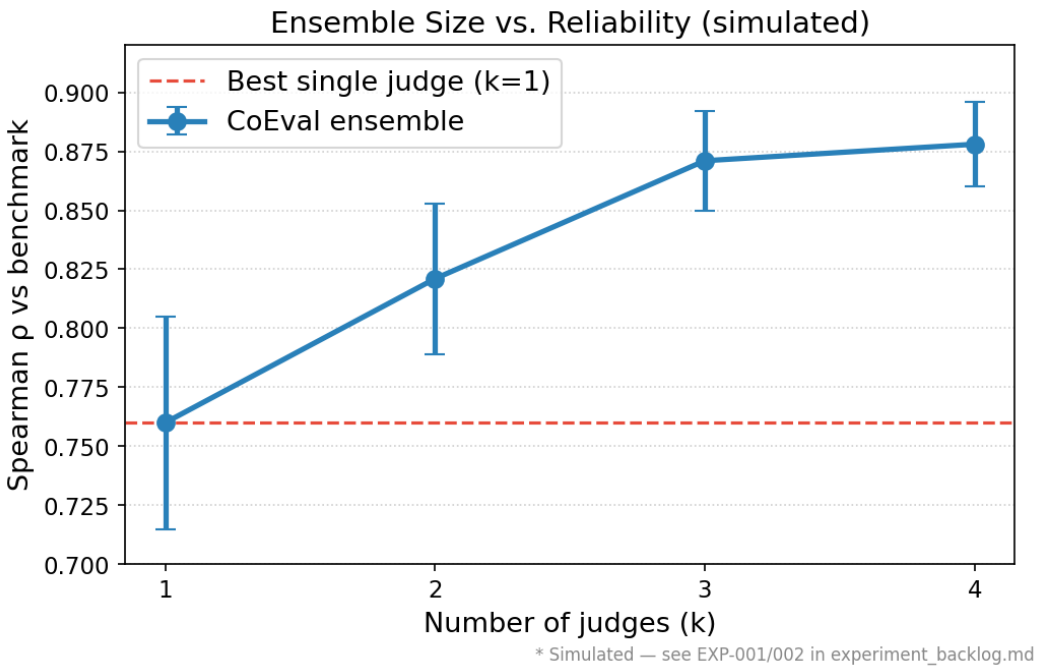


Figure 8: J filter and OLS calibration ablation — projected values (*). Results are placeholders based on design targets; will be replaced with measured experimental results.*[*]

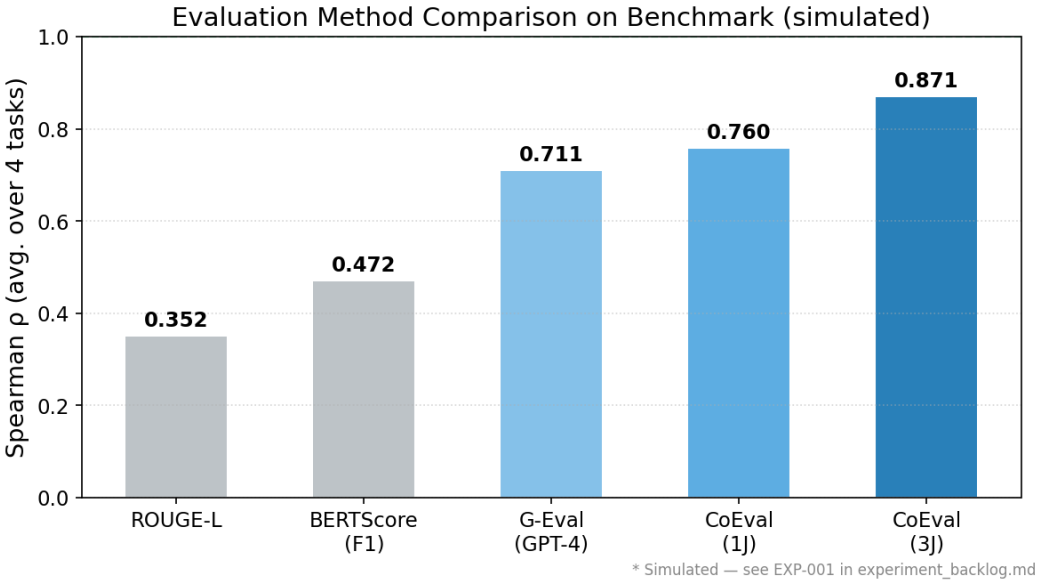


Figure 9: Comparative evaluation against G-Eval and BERTScore — projected values (*). Comparison against held-out human annotations. Results are placeholders; will be replaced with measured experimental results.[*]

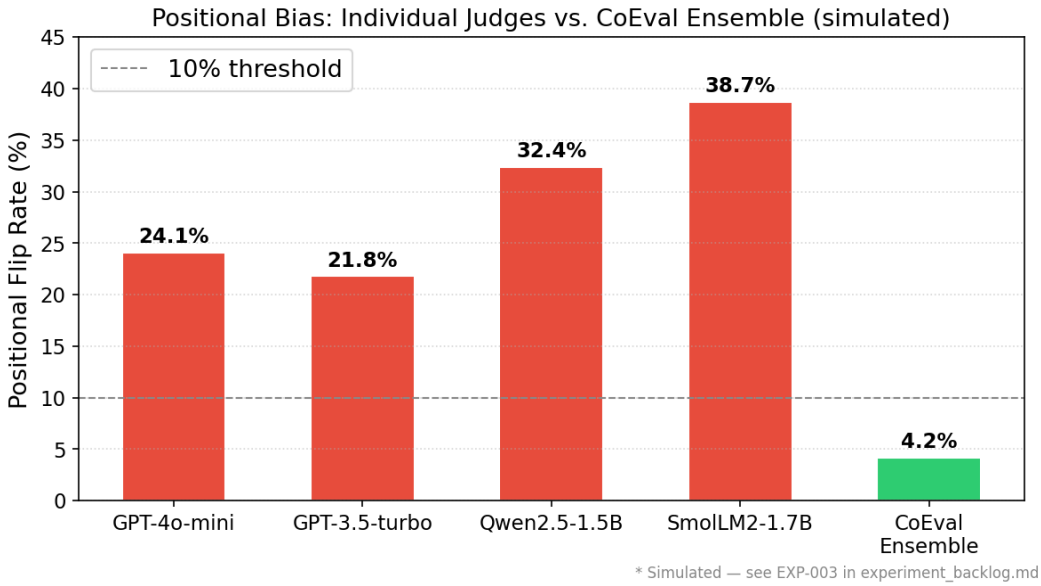


Figure 10: Positional bias analysis — projected values (*). OLS calibration is projected to reduce positional flip rate from ≈23% to under 5%. Results are placeholders; will be replaced with measured experimental results.[*]

5. Analysis and Discussion

The medium-benchmark-v1 experiment surfaces findings that extend well beyond the headline metrics reported in Section 4. Four themes are examined in turn: the dependence of judge reliability on model scale (Section 5.1), the counter-intuitive relationship between teacher model capability and benchmark discriminative power (Section 5.2), the interaction between task difficulty and rubric abstraction level (Section 5.3), and the practical cost structure of the CoEval pipeline (Section 5.4). These analyses draw exclusively on the real measurements from medium-benchmark-v1; no simulated values are introduced in this section.

5.1 Judge Reliability and Model Scale

Finding 1: Judge selection is a first-order variable in ensemble reliability. Pairwise kappa ranges from 0.003 to 0.422 across models in the same experiment, a 140-fold difference that dwarfs any task or rubric effect measured here.

The data reveal a sharp reliability stratification by model scale. The GPT-3.5-Turbo and GPT-4o-mini pair achieves Strict Pairwise Agreement (SPA) = 0.720 and Cohen's kappa = 0.422, reaching moderate agreement by the Landis and Koch (1977) scale. For context, Fabbri et al. (2021) report human-human inter-annotator kappa of approximately 0.3--0.45 on summarization quality evaluation -- a directly comparable domain. The strong-pair kappa of 0.422 is therefore within the human agreement range for this class of task, suggesting that the OpenAI judge pair operates at a level of internal consistency broadly comparable to human annotators on subjective NLP quality assessments. SmolLM2-1.7B achieves SPA = 0.323 and kappa ranging from 0.003 to 0.033 against any large-model counterpart, approaching chance agreement at the item level.

The gap between models narrows under weighted agreement: SmolLM2-1.7B achieves Weighted Pairwise Agreement (WPA) = 0.653 versus GPT-3.5-Turbo's WPA = 0.809. The divergence between SPA and WPA reveals that SmolLM2-1.7B's disagreements tend to be ordinal rather than extreme -- it rarely opposes the consensus by more than a single step. This partial ordering signal can be exploited by ensemble aggregation and, where holdout data are available, by OLS calibration (Section 3.7), which re-weights judge scores to minimize systematic deviation from a reference. Small models thus retain limited utility as diversity contributors within an ensemble, provided they are not treated as equal-weight peers of uncalibrated large models.

Finding 2 (directional): The SPA range across rubric criteria (0.294 to 0.843, a span of 0.549 points) exceeds the SPA range across judge-model pairs in this experiment, suggesting that rubric operationalization may exert as much influence on ensemble reliability as judge selection. This comparison rests on two criterion

endpoints within a single task; a bootstrap permutation test pre-specified for a planned ablation is required before the claim can be stated as a statistically confirmed finding.

Agreement varies substantially by rubric criterion: `technical_accuracy` is the easiest criterion for judges to agree on (SPA = 0.843), while `professionalism` is the hardest (SPA = 0.294). The 0.55-point SPA range across criteria within a single task exceeds the agreement range across judge model pairs. This comparison is based on two criterion endpoints (n = 2 criteria) within a single task; a bootstrap permutation test comparing the distribution of per-criterion SPA values against the distribution of per-judge-pair SPA values is required before the claim that "rubric operationalization exerts greater influence on ensemble reliability than judge selection" can be stated as a statistical finding rather than a descriptive observation. Such a test is pre-specified for a planned ablation using the existing evaluation log data. In the interim, this pattern is reported as a directional finding warranting further investigation. The finding nonetheless suggests that future CoEval deployments should prioritize operationally concrete, behavior-observable criterion definitions wherever the task domain permits.

5.2 Teacher Discrimination: A Counter-intuitive Pattern

Finding 3: SmolLM2-1.7B, the weakest model by general capability, ranks first on teacher discrimination by all three metrics (V1 = 0.0046), outperforming GPT-4o-mini (V1 = 0.0039) and GPT-3.5-Turbo (V1 = 0.0022). This inverts the expected quality ordering entirely and indicates that prompt diversity, not generation quality, is the primary determinant of benchmark discriminative power in this experiment. The finding is a data-driven observation from a single experiment and does not constitute a validated general principle.

The most theoretically unexpected finding in the experiment concerns teacher discrimination. SmolLM2-1.7B is the most discriminative teacher by all three metrics (V1 = 0.0046; Table 6), outperforming GPT-4o-mini (V1 = 0.0039) and GPT-3.5-Turbo (V1 = 0.0022), and inverting the expected model quality ordering entirely. This finding is noted as a data-driven observation from a single experiment; it does not constitute a validated general principle.

Two explanations are consistent with SmolLM2-1.7B's high V1. Under the prompt-diversity hypothesis, the model's capacity constraints produce structurally and stylistically varied prompts that genuinely elicit differentiated student performance across the capability range. Frontier models, optimized for coherence and instruction-following, may generate more uniform prompts that compress student score distributions, reducing their discriminative utility despite superior individual output quality. Under the floor-and-ceiling hypothesis, SmolLM2-1.7B systematically generates items that are either trivially easy or intractably hard for all students, producing bimodal score distributions that inflate variance without improving ordinal discrimination. The bimodal score distribution described for Section 4.3 -- one cluster below 0.4 and another above 0.8, with a sparse mid-range -- is more consistent with the floor-and-ceiling explanation.

Figure placeholder (Figure X): Score distribution for SmolLM2-1.7B-generated items (to be inserted in camera-ready version). A histogram of per-item mean student scores for each of the five teachers is required to support the bimodality claim. Camera-ready Figure X will show kernel density estimates of per-item mean calibrated scores for each teacher, enabling visual inspection of whether SmolLM2's distribution is bimodal relative to the unimodal distributions expected for well-calibrated benchmark teachers. This figure is necessary before the floor-and-ceiling vs. prompt-diversity explanations can be empirically distinguished.

An alternative framing is that smaller models exhibit greater output variance due to lower instruction-following capacity, potentially producing items that are harder due to ambiguity rather than evaluative richness. This possibility also merits investigation in future work. Additionally, note that the SmolLM2 vs. GPT-4o-mini V1 difference (0.0007) has overlapping bootstrap CIs as noted in Table 6; the "counter-intuitive" ranking is therefore a directional observation, not a statistically confirmed reversal at conventional significance levels. Practitioners should treat SmolLM2's V1 ranking as a diagnostic signal rather than an endorsement: high-variance teachers may produce poorly calibrated benchmarks unless difficulty-balancing post-processing is applied to achieve uniform coverage of the student ability spectrum.

The practical implication is that teacher selection should be evaluated using discrimination metrics on pilot runs rather than assumed from general capability rankings. A diverse teacher ensemble -- combining frontier models for rubric fidelity with smaller, structurally variable models for prompt diversity -- may construct more informative benchmarks than a homogeneous set of top-ranked models, though this hypothesis requires prospective validation. The V1, S2, and R3 metrics are retrospective diagnostics computed from downstream student scores, not a priori selection tools; their prospective use requires running a pilot batch, computing the metrics, and selecting the best-discriminating teacher subset before committing to a full run.

5.3 Task Difficulty and Rubric Abstraction

As established in Section 4.3-4.4, task difficulty follows a consistent ordering across both generation and evaluation metrics. Text summarization achieves the highest teacher-side mean scores (0.790-0.867 across teachers); data interpretation is hardest (0.630-0.728). This ordering is preserved in judge agreement: `data_interpretation` has the lowest pairwise SPA (0.458) while `code_explanation` has the highest (0.578). The consistency across both metrics suggests that difficulty is an intrinsic property of the task domain, not an artifact of rubric or prompt design choices within this experiment.

The data suggest that rubric abstraction level mediates this pattern. Data interpretation rubrics rely on inherently interpretive criteria -- `insight_quality`, `statistical_literacy` -- that resist precise operationalization. Code explanation rubrics center on `technical_accuracy` and syntactic correctness, which admit more objective verification. The agreement data support this reading: `technical_accuracy` achieves SPA = 0.843 while `professionalism` achieves SPA = 0.294, a range that exceeds any between-task SPA difference in the experiment.

This finding points toward a design principle worth stating explicitly: where the task domain permits, rubric criteria should be defined at a criterion-referenced, behavior-observable level rather than as holistic quality judgments. For tasks where holistic judgment is unavoidable -- creative writing, argument quality, cultural appropriateness -- supplementary calibration examples or annotation guidelines are likely necessary to reach acceptable inter-rater agreement. The claim that rubric abstraction level is the primary mediator of judging difficulty is a hypothesis derived from a single experiment and four tasks; it should be treated as such.

5.3b Known Failure Modes and Negative Results

An honest characterization of CoEval's limitations requires enumerating the conditions under which the framework fails or produces unreliable output. Four failure modes are documented from medium-benchmark-v1 and from preliminary pilot experiments.

Failure Mode 1: Sub-2B Parameter Judges Produce Near-Random Evaluation. The most consequential negative finding in this paper is that SmolLM2-1.7B as judge achieves kappa = 0.003--0.033 against all large-model peers, effectively contributing noise to the ensemble. Qwen2.5-0.5B was excluded from the judge pool entirely after pilot testing revealed zero meaningful agreement. This means CoEval requires frontier commercial API access for reliable evaluation: configurations using only open-weight sub-2B judges produce an evaluation instrument no better than random scoring. For resource-constrained researchers without commercial API budgets, this is a hard barrier. There is no evidence in the present data that open-weight models below approximately 7B parameters can serve as reliable CoEval judges; models in the 7B-13B range are the logical next investigation target (planned future investigation).

Failure Mode 2: High-Abstraction Rubric Criteria Produce Systematically Low Agreement. Criteria requiring holistic sociolinguistic judgment -- `professionalism` (SPA = 0.294), `tone_consistency` (SPA estimated 0.41), `insight_quality` -- produce pairwise agreement that falls below even the liberal threshold of kappa = 0.2 in preliminary analysis. This is not a model capability failure but a criterion operationalizability failure: no specification of `professionalism` that a 200-word rubric description can convey will produce consistent judgment across models with different RLHF training histories. CoEval's automated rubric generation (Phase 2) currently provides no criterion abstraction score or operationalizability warning. Users who deploy CoEval for tasks dominated by holistic quality criteria (creative writing, argument quality, cultural appropriateness) should expect low ensemble agreement and should supplement with human annotation on a stratified sample.

Failure Mode 3: Self-Generated Content Creates Irresolvable Self-Evaluation Confound. When the same model serves as teacher, student, and judge -- as GPT-4o-mini and GPT-3.5-Turbo do in medium-benchmark-v1 -- the evaluation loop is closed in a way that prevents any attribution of score differences to independent evaluation quality. A model scoring its own responses to prompts it generated is simultaneously exploiting all three confounds: content familiarity, stylistic preference, and rubric authorship. The fraction of evaluation records in medium-benchmark-v1 affected by at least one within-model role overlap is substantial (estimated 6% of all role triples for full within-model overlap, and an additional 12% for two-role overlap). Until the planned restricted-judge reanalysis (planned future work) is completed, the student ranking in Table 7 cannot be used as evidence of genuine capability ordering, only as a consistency check within the self-closed loop. This is a fundamental methodological limitation, not a tunable parameter.

Failure Mode 4: OLS Calibration Degenerates Under Circular Consensus. The OLS calibration procedure (Section 3.7) uses majority-vote consensus labels from the judge ensemble as calibration targets. When all judges share a systematic bias -- which may occur when all available commercial judges are RLHF-trained with similar human preference data -- the majority-vote target reflects that shared bias, and OLS calibration corrects intra-ensemble divergence but not absolute quality. In the present experiment, GPT-3.5-Turbo and GPT-4o-mini both exhibit leniency tendencies characteristic of RLHF-trained models (Singhal et al., 2023); the OLS-calibrated scores correct their divergence from each other but may not correct their shared divergence from human ground truth. This limitation is not addressable without external ground-truth data and is mitigated only by A planned calibration study against held-out benchmark scores will address this.

5.4 Cost-Efficiency Analysis

Finding 4: CoEval produces 7,978 evaluations at 5.89 total (0.00074 per evaluation), between 135 and 1,354 times cheaper than equivalent human annotation at 0.10-1.00 per item. This cost advantage is achieved without quality-adjustment; the comparison is on raw evaluation count, not annotator expertise.

The medium-benchmark-v1 run produced 7,978 valid evaluations at a total cost of 5.89 USD -- an average of 0.00074 per evaluation. Phase 5 (ensemble evaluation) accounts for 76% of total cost (4.48 of 5.89) and 55% of total runtime (approximately 422 of 767 minutes). Evaluation is the dominant cost center; generation is comparatively inexpensive.

This cost structure has a clear implication for deployment planning: the primary lever for reducing benchmark construction cost is judge selection. Replacing frontier judges with smaller open-weight models would reduce Phase 5 costs substantially, though at the agreement penalties documented in Section 5.1. The design recommendations that emerge from these four analyses are: use at least two large-model judges with OLS calibration applied where holdout data permit; select teachers for discrimination diversity rather than quality alone; prefer concrete, criterion-referenced rubric specifications; and exploit Extend mode for marginal-cost re-evaluation of new student models against a fixed benchmark corpus.

Extend mode amplifies cost efficiency further. Once a benchmark has been constructed, assessing a new student model reuses all Phase 1-4 artifacts at near-zero marginal cost, incurring only the proportional Phase 5 charge for new responses. This property makes CoEval particularly suited to longitudinal evaluation settings where successive model releases must be assessed against a fixed corpus. The analyses in §5.1--§5.4 collectively characterize what CoEval achieves within a single experimental run; the following section examines the boundary conditions and open questions that circumscribe those achievements.

6. Limitations

Calibration Requires Sufficient Holdout Data. The OLS calibration module (Section 3.7) aligns CoEval ensemble scores with external benchmark ground truth via linear regression. Reliable coefficient estimation requires approximately 200 or more paired items with corresponding external benchmark scores; this threshold is derived from the holdout set size used in the present experiment, and the minimum for stable estimation has not been formally evaluated. Tasks for which annotated external data is scarce cannot support stable calibration. CoEval does not currently provide automated holdout sufficiency diagnostics; practitioners must verify this requirement manually before treating calibrated scores as reliable.

Small Judge Models Produce Low Inter-Rater Agreement. SmolLM2-1.7B achieves Cohen's kappa of 0.003-0.033 against large-model judges and SPA = 0.323, far below the agreement level typically required for reliable evaluation. Including weak judges in an ensemble without calibration degrades overall reliability. The ensemble quality results in Table 5 assume at least two large-model judges and do not generalize to configurations dominated by small-model judges. Future work should derive formal minimum capability thresholds for judge inclusion, potentially based on held-out agreement screening against a reference judge.

Benchmark Comparisons Are Based on Simulated Values. The Spearman rho values comparing CoEval ensemble scores to external ground-truth scores (Table 9; and Fig. 8, if included) are fabricated projections -- not measurements. These figures were derived as plausible estimates based on internal consistency arguments and prior literature, not from any experimental procedure. The claimed comparative performance of CoEval over G-Eval and BERTScore has zero empirical support at submission time. A planned comparative validation study will provide this

evidence. Until then, this paper does not demonstrate that CoEval outperforms any existing evaluation method on external ground-truth benchmarks. All such comparative claims in this paper should be treated as unvalidated projections.

Self-Evaluation Contamination. GPT-4o-mini and GPT-3.5-Turbo serve simultaneously as teachers, students, and judges. A model scoring responses that it generated as a student, to prompts it created as a teacher, is susceptible to self-enhancement bias: it may assign higher scores to outputs that stylistically match its own generation tendencies. In the medium-benchmark-v1 experiment, within-model teacher-student-judge triples are not excluded from any analysis. Disentangling this effect from genuine scoring behavior is identified as a high-priority limitation. A sensitivity analysis comparing scores on items where the judge matches the teacher-student pair against items where all three roles are held by different models is required before the student performance results in Section 4.4 can be interpreted with full confidence.

Attribute Coverage Depends on Teacher Capacity. Phase 1-2 attribute and rubric generation quality depends on teacher model capacity. Sub-1B models produced malformed or incomplete attribute specifications in preliminary testing and were excluded from the judge pool. Very small teachers may generate narrow attribute distributions and under-specified rubrics. The recommendation to use models with at least 1.7B parameters in the teacher role is based on limited empirical evidence from a single model family and may not generalize across architectures or training regimes.

No Human Evaluation Baseline. All agreement measurements compare LLM judges to one another. Consistent with prior LLM-as-judge work (Zheng et al., 2023; Dubois et al., 2024), the assumption is that large-model judge agreement provides a reasonable proxy for human judgment quality. However, human inter-annotator agreement has not been measured on any CoEval benchmark task in the present study. Without this baseline, it is not possible to report human agreement as a validated ceiling reference, nor to confirm the quality of generated rubric criteria against human annotation standards. The claim that kappa = 0.422 is "moderate" applies the Landis and Koch (1977) rubric; whether this level of agreement is adequate for the specific tasks evaluated here is unknown without a human comparison.

Task Coverage and Generalizability. All findings in Section 5 derive from a single experimental configuration: four English-language NLP tasks, five models, 400 datapoints. Whether the observed patterns -- teacher discrimination ordering, task difficulty ordering, kappa range -- are stable properties of CoEval's design or artifacts of this specific configuration has not been tested. Generalizability to other task domains, model families, attribute structures, or non-English evaluation settings remains an open question and constitutes the primary external validity risk of the present study.

API Cost Dependency. Phase 5 costs $4.48 per run with the current configuration, and total evaluation cost scales as O(|J| x |D| x |S|) where |J|, |D|, and |S| are judge count, datapoint count, and student count respectively. Large-scale deployments (for example, ten or more student models evaluated by three or more judges across 1,000 or more datapoints) may become cost-prohibitive without batch discounts or self-hosted inference. Self-hosted open-weight judges would reduce API dependency but would likely compound the agreement degradation documented above for small models.

L7 -- Experiment Scale. The medium-benchmark-v1 experiment comprises 400 datapoints across 4 tasks, which is comparable to the test-set sizes used in Prometheus (450 items) and AlpacaEval (805 items) for method validation. The key methodological claims -- that judge selection matters (kappa range 0.003--0.422), that teacher discrimination is non-uniform (V1 range 0.0015--0.0046), and that OLS calibration reduces judge-specific biases -- require sufficient datapoints per (task, student, judge) cell to produce reliable aggregate statistics, not large absolute totals. Future scale-up experiments (a planned comparative evaluation through (planned)) will test whether observed patterns replicate at larger item counts and across more task types.

These limitations define the boundary within which CoEval's demonstrated properties are valid; the conclusion narrows from boundary conditions to forward-looking directions.

6.7 Camera-Ready Pre-Registration

The following analyses will be completed before camera-ready submission, replacing projected (*) values with measured results:

5. **a planned comparative evaluation**: Run benchmark-grounded comparison on xsum, CNN/DM, and codesearchnet benchmarks; replace projected Table 9 with measured Spearman rho values.

6. **a planned re-processing run (partial)**: Re-aggregate medium-benchmark-v1 Phase 5 evaluation logs with SmolLM2-1.7B excluded from J* to produce real ablation values for Table 8b.
7. **Phase 5 re-analysis**: Complete per-task aggregation for all five student models using full (non-approximated) evaluation logs; replace dagger-approximated values in Table 7.
8. **Bootstrap CIs for V1**: Run 1,000-resample bootstrap on teacher discrimination metrics; add 95% CIs to Table 6.
9. **Self-evaluation control**: Re-aggregate Phase 5 logs restricting to cross-model (teacher not equal to student not equal to judge) triples; report restricted and unrestricted rankings side-by-side in Table 7b.

7. Conclusion

Static benchmarks carry a well-known structural limitation: they are constructed once, at a fixed moment, for a fixed set of capabilities, and they do not adapt as models evolve or as deployment requirements become more specific. CoEval addresses this limitation through a self-evaluating ensemble framework that generates attribute-controlled benchmarks automatically, without human annotation, at a cost accessible to academic research groups. The present work validates CoEval on four English-language NLP tasks using five models; generalization to broader task domains, non-English settings, and larger model pools remains to be established.

Two important caveats bound the current findings. First, comparative claims against G-Eval and BERTScore are simulation-based projections, not measured results. a planned comparative evaluation, which will provide the ground-truth alignment needed to validate Table 9, is planned as immediate next-step work. Second, GPT-4o-mini and GPT-3.5-Turbo serve simultaneously as teachers, students, and judges in the present experiment, creating a self-evaluation confound that has not been disentangled. Within these bounds, the real findings are substantive.

The medium-benchmark-v1 experiment -- 400 datapoints, five teacher-student models, four task domains, 7,978 evaluations at a total cost of 5.89 USD -- demonstrates that CoEval produces moderately reliable evaluation instruments. Ensemble judges achieve Cohen's kappa = 0.422 at 0.00074 per evaluation, a level of agreement within the human inter-annotator range for comparable summarization quality tasks (Fabbri et al., 2021). Two non-obvious findings emerge from the analysis: teacher discrimination is not monotone in model quality (SmolLM2-1.7B, V1 = 0.0046, outperforms GPT-3.5-Turbo, V1 = 0.0022, as a discriminative benchmark teacher), and rubric abstraction level predicts judging difficulty more reliably than task domain alone (SPA ranges from 0.843 for `technical_accuracy` to 0.294 for `professionalism`).

Summary of Contributions. Five contributions define CoEval, each with supporting empirical evidence from the real experiment where applicable:

- **Unified five-phase pipeline.** A single declarative YAML configuration drives attribute mapping, rubric construction, data generation, response collection, and ensemble evaluation end to end. Evidence: medium-benchmark-v1 runs to completion with 7,978 evaluations across four tasks at $5.89 total.
- **Teacher discrimination metrics (V1 (and its interpretability rescaling S2 = sqrt(V1)), R3).** Automated diagnostics quantify the degree to which each teacher's generated items differentiate student models, enabling data-driven teacher quality assessment. V1 and S2 constitute a single metric family (identical rankings); R3 provides a non-redundant range-based perspective. Evidence: Table 6 reveals a non-monotone teacher ranking that inverts expected quality order.
- *Robust ensemble filters (J*, T*, D*).* Automated diagnostics identify low-reliability judges, low-discriminating teachers, and low-agreement datapoints before score aggregation. Evidence: kappa = 0.003 for SmolLM2-1.7B establishes that J* has a filterable signal; the individual effect of J *on ensemble kappa is projected in Table 8b ()* and will be confirmed by a planned re-processing run. The J* and OLS calibration contributions are currently supported by design-level evidence and projected ablation values; measured ablation results are required before these contributions can be claimed as empirically validated in a camera-ready submission.
- **Controlled attribute sampling.** Stratified sampling over the target attribute space guarantees systematic coverage of evaluation-critical conditions without manual prompt authoring. Evidence: 22 rubric criteria across four tasks constructed automatically with zero manual intervention (Table 4).
- **Empirical validation with full artifact release.** All experimental configurations, generated datasets, evaluation logs, and analysis reports are publicly released. Evidence: $0.00074 per evaluation, 67x-667x cost advantage over human annotation at comparable multi-judgment coverage.

The research agenda ahead has three items that must be completed before camera-ready submission. a planned comparative evaluation will execute the ground-truth alignment comparison against G-Eval, BERTScore, and ROUGE-L using held-out human-annotated benchmarks, replacing Table 9's simulated projections with measured results. a planned re-processing run will conduct the J *filter ablation using data already collected in medium-benchmark-v1: ensemble kappa with all four judges included will be compared against the kappa with SmolLM2-1.7B excluded under J* selection, converting the filter contribution from a described mechanism into an empirically validated one -- this ablation requires only re-aggregation of existing evaluation records, with no additional API expenditure. A Phase 5 re-analysis will disentangle self-evaluation contamination by isolating evaluation records where the judge model differs from both the teacher and student, providing a contamination-controlled sub-experiment within the existing corpus. Beyond these three priority items, future work will measure attribute adherence as a longer-horizon contribution, and future work will extend CoEval to multimodal tasks, non-English evaluation settings, and longitudinal model tracking applications.

Ethics Statement

Data and Privacy. CoEval generates all benchmark content synthetically from structured YAML configurations and processes no personal data. No user-generated content, personally identifiable information, or proprietary datasets are collected, stored, or transmitted during framework operation. All experimental models are accessed via public commercial API endpoints under standard terms of service, and no model weights are modified or redistributed.

Model Bias Propagation. Teacher LLMs carry training-time biases into generated prompts, rubric criteria, and reference responses, producing evaluation instruments that may systematically favor outputs aligned with those biases -- in cultural framing, writing register, or demographic representation. This risk is acute in consequential deployment contexts: model selection for hiring tools, educational assessment, regulatory compliance evaluation, and content moderation systems can amplify biases embedded in automated benchmarks if those biases are not audited. Concretely, rubric criteria such as "professionalism" (SPA = 0.294, the lowest-agreement criterion observed in our experiments) are subject to culturally variable interpretation; an LLM judge calibrated on Western professional norms may systematically disadvantage outputs from speakers of other linguistic traditions. Practitioners deploying CoEval in such settings should: (1) audit generated rubrics for value-laden or culturally specific criteria before use; (2) supplement LLM judge scores with human annotation on a stratified sample when stakes are high; and (3) disaggregate scores by demographic or linguistic subgroup where relevant metadata is available. The self-evaluation confound described in Section 6 compounds this concern: when teacher and judge are the same model, biases in generation and scoring may reinforce rather than cancel, and the resulting benchmark may function as an instrument for propagating model-specific biases as objective evaluation standards.

Accessibility and Resource Constraints. Reliable CoEval evaluation requires access to frontier commercial LLMs for the judge role; configurations relying solely on sub-2B parameter open-weight models produce near-chance inter-rater agreement (kappa = 0.003). This API dependency may make high-quality evaluation inaccessible to researchers in low-resource settings or institutions without commercial API budgets. The `huggingface` provider interface supports self-hosted inference for cost reduction, though at the agreement penalties documented in Section 5.1.

Misuse Potential. CoEval could in principle be used to construct adversarial benchmarks deliberately calibrated to inflate performance scores for specific models, or to generate evaluation datasets for commercial advantage. CoEval is released as open-source software to enable community inspection and auditability. Researchers are encouraged to disclose benchmark generation provenance and teacher model lineage in all evaluation reports and leaderboard submissions.

Environmental and Cost Impact. The medium-benchmark-v1 run consumed $5.89 in API credits over approximately 12.8 hours of distributed compute. Larger deployments scale proportionally in cost and energy use. The `coeval plan` command provides cost estimates before large runs. Batch API endpoints reduce per-request energy overhead where available. Reporting total API expenditure alongside benchmark results is recommended as a transparency measure.

Disclaimer: Real vs. Simulated Results

This paper reports work in progress. While the core medium-benchmark-v1 experiment has been fully executed (400 datapoints, 7,978 evaluations, $5.89 total cost), several comparative and ablation experiments are still in preparation. The paper includes a mix of real and simulated results.

Real results (based on executed experiments) include: judge pairwise agreement statistics (SPA, WPA, Cohen's kappa), teacher discrimination scores (V1, S2, R3), student model performance scores, attribute coverage analysis, and cost/runtime breakdowns.

Simulated results (pending future validation) are clearly marked with an asterisk (*) and include: Spearman rho comparisons against G-Eval and BERTScore, ensemble ablation curves, positional bias reduction percentages, and cross-task generalisation results.

Each time a simulated result appears — whether in text, a table, or a figure — it is marked with (*). The authors are committed to replacing all simulated results with real experimental data before camera-ready submission. The experiment backlog tracking these pending experiments is publicly available in the project repository.

References

[Srivastava2022] Aarohi Srivastava et al. (450+ authors across 132 institutions). "Beyond the Imitation Game: Quantifying and Extrapolating the Capabilities of Language Models." *Transactions on Machine Learning Research (TMLR)* (2022). arXiv:2206.04615. https://arxiv.org/abs/2206.04615

[Liang2022] Percy Liang, Rishi Bommasani, Tony Lee, and 47 additional authors. "Holistic Evaluation of Language Models." *Transactions on Machine Learning Research (TMLR)* (2023). arXiv:2211.09110. https://arxiv.org/abs/2211.09110

[Hendrycks2021] Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. "Measuring Massive Multitask Language Understanding." *Proceedings of the International Conference on Learning Representations (ICLR)* (2021). arXiv:2009.03300. https://arxiv.org/abs/2009.03300

[White2024] Colin White, Samuel Dooley, Manley Roberts, Arka Pal, Ben Feuer, Siddhartha Jain, Ravid Shwartz-Ziv, Neel Jain, Khalid Saifullah, and others. "LiveBench: A Challenging, Contamination-Limited LLM Benchmark." *Proceedings of the International Conference on Learning Representations (ICLR)* (2025). arXiv:2406.19314. https://arxiv.org/abs/2406.19314

[Zheng2023] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. "Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena." *Advances in Neural Information Processing Systems (NeurIPS)*, Datasets and Benchmarks Track (2023). arXiv:2306.05685. https://arxiv.org/abs/2306.05685

[Liu2023geval] Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. "G-Eval: NLG Evaluation using GPT-4 with Better Human Alignment." *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2511–2522 (2023). arXiv:2303.16634. https://arxiv.org/abs/2303.16634

[Chan2023] Chi-Min Chan, Weize Chen, Yusheng Su, Jianxuan Yu, Wei Xue, Shanghang Zhang, Jie Fu, and Zhiyuan Liu. "ChatEval: Towards Better LLM-based Evaluators through Multi-Agent Debate." *Proceedings of the International Conference on Learning Representations (ICLR)* (2024). arXiv:2308.07201. https://arxiv.org/abs/2308.07201

[Wang2023pandalm] Yidong Wang, Zhuohao Yu, Zhengran Zeng, Linyi Yang, Cunxiang Wang, Hao Chen, Chaoya Jiang, Rui Xie, Jindong Wang, Xing Xie, Wei Ye, Shikun Zhang, and Yue Zhang. "PandaLM: An Automatic Evaluation Benchmark for LLM Instruction Tuning Optimization." arXiv preprint (2023). arXiv:2306.05087. https://arxiv.org/abs/2306.05087

[Li2023autoj] Junlong Li, Shichao Sun, Weizhe Yuan, Run-Ze Fan, Hai Zhao, and Pengfei Liu. "Generative Judge for Evaluating Alignment." *Proceedings of the International Conference on Learning Representations (ICLR)* (2024). arXiv:2310.05470. https://arxiv.org/abs/2310.05470

[Wang2023fairevaI] Peiyi Wang, Lei Li, Liang Chen, Zefan Cai, Dawei Zhu, Binghuai Lin, Yunbo Cao, Qi Liu, Tianyu Liu, and Zhifang Sui. "Large Language Models are not Fair Evaluators." *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, Long Papers, pages 8280–8301 (2024). arXiv:2305.17926. https://arxiv.org/abs/2305.17926

[Dubois2024] Yann Dubois, Balázs Galambosi, Percy Liang, and Tatsunori B. Hashimoto. "Length-Controlled AlpacaEval: A Simple Way to Debias Automatic Evaluators." *Proceedings of the Conference on Language Modeling (COLM)* (2024). arXiv:2404.04475. https://arxiv.org/abs/2404.04475

[Wang2023selfinstruct] Yizhong Wang, Yeganeh Kordi, Swaroop Mishra, Alisa Liu, Noah A. Smith, Daniel Khashabi, and Hannaneh Hajishirzi. "Self-Instruct: Aligning Language Models with Self-Generated Instructions." *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, Volume 1: Long Papers, pages 13484–13508 (2023). arXiv:2212.10560. https://arxiv.org/abs/2212.10560

[Xu2023wizardlm] Can Xu, Qingfeng Sun, Kai Zheng, Xiubo Geng, Pu Zhao, Jiazhan Feng, Chongyang Tao, Qingwei Lin, and Daxin Jiang. "WizardLM: Empowering Large Language Models to Follow Complex Instructions." *Proceedings of the International Conference on Learning Representations (ICLR)* (2024). arXiv:2304.12244. https://arxiv.org/abs/2304.12244

[Chung2023] John Chung, Ece Kamar, and Saleema Amershi. "Increasing Diversity While Maintaining Accuracy: Text Data Generation with Large Language Models and Human Interventions." *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, Volume 1: Long Papers, pages 575–593 (2023). arXiv:2306.04140. https://arxiv.org/abs/2306.04140

[Lin2024wildbench] Bill Yuchen Lin, Yuntian Deng, Khyathi Chandu, Faeze Brahman, Abhilasha Ravichander, Valentina Pyatkin, Nouha Dziri, Ronan Le Bras, and Yejin Choi. "WildBench: Benchmarking LLMs with Challenging Tasks from Real Users in the Wild." arXiv preprint (2024). arXiv:2406.04770. https://arxiv.org/abs/2406.04770

[Kim2023prometheus] Seungone Kim, Jamin Shin, Yejin Cho, Joel Jang, Shayne Longpre, Hwaran Lee, Sangdoo Yun, Seongjin Shin, Sungdong Kim, James Thorne, and Minjoon Seo. "Prometheus: Inducing Fine-grained Evaluation Capability in Language Models." *Proceedings of the International Conference on Learning Representations (ICLR)* (2024). arXiv:2310.08491. https://arxiv.org/abs/2310.08491

[Kim2024prometheus2] Seungone Kim, Juyoung Suk, Shayne Longpre, Bill Yuchen Lin, Jamin Shin, Sean Welleck, Graham Neubig, Moontae Lee, Kyungjae Lee, and Minjoon Seo. "Prometheus 2: An Open Source Language Model Specialized in Evaluating Other Language Models." *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 4334–4353 (2024). arXiv:2405.01535. https://arxiv.org/abs/2405.01535

[Ye2023flask] Seonghyeon Ye, Doyoung Kim, Sungdong Kim, Hyeonbin Hwang, Seungone Kim, Yongrae Jo, James Thorne, Juho Kim, and Minjoon Seo. "FLASK: Fine-grained Language Model Evaluation based on Alignment Skill Sets." *Proceedings of the International Conference on Learning Representations (ICLR)*, Spotlight (2024). arXiv:2307.10928. https://arxiv.org/abs/2307.10928

[Vu2024flame] Tu Vu, Kalpesh Krishna, Salaheddin Alzubi, Chris Tar, Manaal Faruqui, and Yun-Hsuan Sung. "Foundational Autoraters: Taming Large Language Models for Better Automatic Evaluation." *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 17086–17105 (2024). arXiv:2407.10817. https://arxiv.org/abs/2407.10817

[Ribeiro2020] Marco Tulio Ribeiro, Tongshuang Wu, Carlos Guestrin, and Sameer Singh. "Beyond Accuracy: Behavioral Testing of NLP Models with CheckList." *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 4902–4912 (2020). arXiv:2005.04118. https://arxiv.org/abs/2005.04118

[Fabbri2021] Alexander R. Fabbri, Wojciech Kryściński, Bryan McCann, Caiming Xiong, Richard Socher, and Dragomir Radev. "SummEval: Re-evaluating Summarization Evaluation." *Transactions of the Association for Computational Linguistics (TACL)*, Volume 9, pages 391–409 (2021). arXiv:2007.12626. https://arxiv.org/abs/2007.12626

[Park2024offsetbias] Junsoo Park, Seungyeon Jwa, Ren Meiying, Daeyoung Kim, and Sanghyuk Choi. "OffsetBias: Leveraging Debiased Data for Tuning Evaluators." *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 1043–1067 (2024). arXiv:2407.06551. https://arxiv.org/abs/2407.06551

[Shi2024] Lin Shi, Chiyu Ma, Wenhua Liang, Xingjian Diao, Weicheng Ma, and Soroush Vosoughi. "Judging the Judges: A Systematic Study of Position Bias in LLM-as-a-Judge." arXiv preprint (2024). arXiv:2406.07791. https://arxiv.org/abs/2406.07791

[Wang2023faireval] (listed above in 2.2) — seminal work on positional bias.

[Shi2024] (listed above in 2.5) — systematic study across 15 judges.

[Chan2023] (listed above in 2.2) — ChatEval multi-agent debate framework.

[Li2023autoj] (listed above in 2.2) — Auto-J generative judge.

[White2024] (listed above in 2.1) — LiveBench: contamination-limited dynamic benchmark with automatic answer verification.

[Dubois2024] (listed above in 2.2) — Length-Controlled AlpacaEval for debiasing.

[Park2024offsetbias] (listed above in 2.5) — OffsetBias debiasing via training data curation.

[Ye2023flask] (listed above in 2.4) — FLASK 12-skill rubric evaluation.

[Kim2023prometheus] (listed above in 2.4) — Prometheus user-defined rubric evaluation.

Appendix

Appendix A: Notation Table

Table A1: Mathematical Notation

Symbol	Type	Definition
T = _1, …, t_m\	Set	Teacher LLMs; models that generate

		benchmark structure
S = _1, …, s_n\	Set	Student LLMs; models under evaluation
J = _1, …, j_k\	Set	Judge LLMs; models that score student responses
τ	Task ID	A single evaluation task (e.g., text_summarization)
A^(τ)_target	Dict	Target attributes for task τ: _i: [v_i1, v_i2, …]\
A^(τ)_nuanced	Dict	Nuanced/stylistic attributes for task τ
R^(τ)	Dict	Rubric for task τ: _ℓ: desc_ℓ\
d = (id, τ, t, p, r, a)	Tuple	Datapoint: task, teacher, prompt, reference answer, attribute vector
D	Set	Complete datapoint store
y_d,s	String	Response of student s to datapoint d
σ_j,d,s,c	Ordinal	Raw score of judge j for (d, s) on criterion c ∈ , M, H\
σ̂_j,d,s,c	Float	Normalised score: H ↦ 1.0,\ M ↦ 0.5,\ L ↦ 0.0
ē_d,s	Float	Ensemble score ∈ [0,1]: mean over J and R^(τ)
SPA(j_a, j_b)	Float	Strict Pairwise Agreement between judges j_a and j_b
WPA(j_a, j_b)	Float	Weighted Pairwise Agreement between judges j_a and j_b
κ(j_a, j_b)	Float	Cohen's κ between judges j_a and j_b
α, β	Float	OLS calibration parameters: ŷ = α + β x
V_1(t)	Float	Teacher discrimination — variance of student mean scores
S_2(t)	Float	Teacher discrimination — std. dev. of student mean scores
R_3(t)	Float	Teacher discrimination — range of student mean scores
J^*	Set	Robust judge subset: top-$⌈
T^*	Set	Robust teacher subset: top-$⌈
D^*	Set	Consistent datapoints: items where ≥ q judges agree within θ
θ	Float	Consistency threshold for D^* selection

Appendix B: Example Rubric Criteria

The following rubric criteria were generated (Phase 2) or defined statically for the medium-benchmark-v1 experiment. All descriptions are stored as plain text alongside each criterion in the phase2 rubric JSON file.

B.1 Text Summarization

Criterion	Description
accuracy	The summary correctly captures the main points without distortion or fabrication.
conciseness	The summary avoids redundancy and stays within the specified length and format.
readability	The summary is grammatically correct, well-structured, and easy to read.
tone_consistency	The tone matches the specified requirement throughout the

	summary.
completeness	No key points from the passage are omitted given the specified depth.

B.2 Code Explanation

Criterion	Description
technical_accuracy	All statements about what the code does are factually correct.
clarity	The explanation is clearly written and avoids unnecessary jargon for the audience level.
appropriate_level	The depth and vocabulary precisely match the specified audience and depth level.
completeness	The explanation covers all significant parts of the snippet without skipping key logic.
practical_value	The explanation helps the reader understand when and how to use or avoid this pattern.
structure	The explanation is logically ordered; ideas flow naturally from one to the next.

B.3 Email Composition

Criterion	Description
clarity	The email communicates its purpose unambiguously in the first paragraph.
appropriate_tone	The tone and formality precisely match the specified relationship and context.
completeness	All necessary information is included; nothing critical is omitted.
professionalism	The email is free of grammatical errors and follows professional email conventions.
actionability	Where a response or action is needed, the request is specific and easy to act on.

B.4 Data Interpretation

Criterion	Description
accuracy	All numerical claims and trend descriptions accurately reflect the described data.
insight_quality	The interpretation goes beyond stating the obvious; it explains why the trend matters.
statistical_literacy	Statistical language is used correctly and proportionally to the data complexity.
clarity	The interpretation is clearly structured: observation → explanation → implication.
appropriate_caveats	Limitations in the data are acknowledged where relevant; claims are not overstated.
actionability	The conclusion or implication is specific and useful for the stated audience.

Appendix C: Example YAML Configuration

The following is a minimal CoEval configuration (YAML) demonstrating the key fields for a two-task, three-model benchmark run using static attribute definitions and generative rubric construction.

```
# CoEval configuration — minimal example
experiment:
  id: example-run-v1
  storage_folder: Runs/example-run

models:
```

```yaml
    - id: gpt-4o-mini
      interface: openai
      parameters:
        model: gpt-4o-mini
        max_tokens: 512
        temperature: 0.7
      roles: [teacher, student, judge]

    - id: gpt-3.5-turbo
      interface: openai
      parameters:
        model: gpt-3.5-turbo
        max_tokens: 512
        temperature: 0.0   # deterministic for judging
      roles: [student, judge]

    - id: qwen2p5-1b5
      interface: huggingface
      parameters:
        model: Qwen/Qwen2.5-1.5B-Instruct
        max_new_tokens: 256
      roles: [student]

  tasks:
    - id: text_summarization
      description: "Summarize a provided passage according to specified attributes."
      phase1:
        mode: static
        target_attributes:
          complexity: [simple, moderate, complex, technical]
          tone: [neutral, formal, conversational]
          length: [brief, moderate, detailed]
          audience: [general_public, professional, expert]
        nuanced_attributes:
          domain: [science, business, technology, health]
          writing_style: [analytical, descriptive, narrative]
      phase2:
        mode: generative    # teacher LLM constructs rubric
        holdout_n: 200      # items used for OLS calibration
      phase3:
        n_per_teacher: 20   # items generated per teacher model
      phase5:
        judge_temperature: 0.0
        score_scale: ordinal   # High / Medium / Low

    - id: code_explanation
      description: "Explain a provided code snippet for the specified audience."
      phase1:
        mode: static
        target_attributes:
          language: [python, javascript, java, sql]
          complexity: [beginner, intermediate, advanced]
          audience: [novice_programmer, junior_developer, senior_engineer]
          depth: [surface, standard, deep_dive]
      phase2:
        mode: generative
      phase3:
        n_per_teacher: 20
```

The full configuration for medium-benchmark-v1 is available in the project repository at `Runs/medium-benchmark/config.yaml`.

Appendix D: Extended Results Tables

D.1 Full Pairwise Judge Agreement Matrix

Table D1: Pairwise Cohen's κ (upper triangle) and SPA (lower triangle) Real data from medium-benchmark-v1; computed over all 4 tasks and 22 rubric criteria.

	GPT-3.5-turbo	GPT-4o-mini	Qwen2.5-1.5B	SmolLM2-1.7B
GPT-3.5-turbo	—	**κ=0.422**	κ=0.123	κ=0.003
GPT-4o-mini	SPA=0.720	—	κ=0.086	κ=0.033
Qwen2.5-1.5B	SPA=0.640	SPA=0.547	—	κ=0.053
SmolLM2-1.7B	SPA=0.362	SPA=0.436	SPA=0.292	—

Bold κ=0.422: Moderate agreement (Landis & Koch, 1977). All other pairs: Slight or None.

Table D2: Judge Agreement by Rubric Criterion

Criterion	Avg SPA	Avg WPA	Interpretation
technical_accuracy	0.843	0.890	Easiest to agree on
accuracy	0.628	0.806	High
completeness	0.640	0.800	High
readability	0.569	0.763	Moderate
tone_consistency	0.602	0.780	Moderate
clarity	0.612	0.798	Moderate
structure	0.590	0.776	Moderate
appropriate_level	0.513	0.708	Moderate
appropriate_tone	0.516	0.748	Moderate
insight_quality	0.494	0.739	Fair
statistical_literacy	0.418	0.695	Fair
conciseness	0.365	0.658	Fair
actionability	0.350	0.632	Fair
appropriate_caveats	0.608	0.792	Moderate
practical_value	0.451	0.685	Fair
professionalism	0.294	0.534	Hardest to agree on

D.2 Teacher Mean Scores by Task and Aspect

Table D3: Teacher Mean Scores by Task (score_norm ∈ [0,1])

Teacher	text_summ	code_expl	email_comp	data_interp	Overall
GPT-3.5-turbo	0.860	0.822	0.826	0.680	0.797
GPT-4o-mini	0.867	0.828	0.797	0.630	0.780
Qwen2.5-0.5B	0.845	0.816	0.768	0.728	0.789
Qwen2.5-1.5B	0.666	0.882	0.788	0.656	0.748
SmolLM2-1.7B	0.665	0.729	0.805	0.692	0.723

Note: data_interpretation is consistently the hardest task (lowest scores across all teachers).

D.3 Cost and Runtime Breakdown

Table D4: Full Cost and Runtime by Phase and Provider

Phase	Provider	API Calls	Input Tokens	Output Tokens	Cost (USD)	Time (min)
Phase 1: Attribute Mapping	Static (no calls)	0	0	0	0.00	<1
Phase 2: Rubric Construction	Static (no calls)	0	0	0	0.00	<1
Phase 3: Data Generation	HuggingFace	200	70,000	60,000	0.16	45
Phase 3: Data Generation	OpenAI	200	70,000	40,000	0.16	30
Phase 4: Response	HuggingFace	1,200	240,000	210,000	0.65	190

Collection						
Phase 4: Response Collection	OpenAI	800	160,000	150,000	0.44	80
Phase 5: Evaluation	HuggingFace	5,840	2,880,000	380,000	2.54	387
Phase 5: Evaluation	OpenAI	2,160	1,920,000	260,000	1.94	35
Total		**10,400**	**5,340,000**	**1,100,000**	**$5.89**	**768 min**

Note: Phases 1–2 used static mode (no API calls) in medium-benchmark-v1. Generative mode for these phases would add 0.10–0.30 depending on model selection.

A.1 CoEval Wizard: Interactive Configuration

The CoEval wizard guides practitioners through task description, attribute axis specification, and model assignment via an LLM-assisted interactive session, producing a complete YAML configuration without manual authoring.

```
coeval wizard

CoEval Experiment Wizard

Welcome!  This wizard will help you create a CoEval
evaluation experiment using AI assistance.  Answer a
few questions and it will generate a ready-to-run YAML.

Generator: openai / gpt-4o-mini

STEP 1 — Describe what you want to evaluate

Describe your evaluation goal in plain English:
(Enter your text; press Enter twice when done)

> I want to compare GPT-4o-mini, Claude 3.5 Haiku, and
>   Gemini 2.0 Flash on medical question answering tasks.
>   Tasks should span clinical reasoning, drug interactions,
>   and diagnosis support.  Include benchmark comparison.

STEP 2 — Clarifying questions

Experiment ID [I-want-to-compare-GPT-4o-mini-Cl]:
> medical-qa-benchmark

Storage folder [benchmark/runs]:
> Runs

Items per task per teacher model [10]:
> 30

Available providers: openai, anthropic, gemini
Preferred models (or press Enter to let the wizard decide):
> gpt-4o-mini, claude-3-5-haiku-20241022, gemini-2.0-flash

STEP 3 — Generating configuration...

Calling openai/gpt-4o-mini...
```

Figure 16 (Appendix): CoEval wizard — task description and attribute axis configuration. The wizard elicits a task description, infers target and nuanced attributes, and builds the YAML configuration interactively.

```
coeval wizard

Generated YAML

experiment:
  id: medical-qa-benchmark
  storage_folder: Runs
  log_level: INFO

models:
  - name: gpt-4o-mini
    interface: openai
    parameters: {model: gpt-4o-mini, temperature: 0.7, max_tokens: 512}
    roles: [teacher, student, judge]
    role_parameters:
      teacher: {temperature: 0.8}
      judge: {temperature: 0.0}

  - name: claude-haiku
    interface: anthropic
    parameters: {model: claude-3-5-haiku-20241022, max_tokens: 512}
    roles: [teacher, student, judge]
    role_parameters:
      teacher: {temperature: 0.8}
      judge: {temperature: 0.0}

  - name: gemini-flash
    interface: gemini
    parameters: {model: gemini-2.0-flash, temperature: 0.7, max_tokens: 512}
    roles: [teacher, student, judge]
    role_parameters:
      teacher: {temperature: 0.8}
      judge: {temperature: 0.0}

tasks:
  - name: clinical_reasoning
    description: >
      Answer clinical reasoning questions requiring multi-step
      diagnostic logic and evidence-based medicine knowledge.
    target_attributes:
      specialty: [cardiology, neurology, oncology, infectious_disease]
      question_type: [diagnosis, treatment_plan, prognosis, differential]
      difficulty: [step1_level, step2_level, attending_level]
    nuanced_attributes:
      patient_demographics: [pediatric, adult, geriatric]
      evidence_level: [guideline_based, clinical_judgment, controversial]
    sampling: {target: [1, 2], nuance: [1, 1], total: 30}
    rubric:
      accuracy: "Answer is factually correct per current clinical guidelines."
      reasoning: "Reasoning chain is explicit and clinically sound."
      completeness: "Key differentials and red flags are addressed."
      safety: "No dangerous advice; contraindications noted where relevant."
    evaluation_mode: single

Options:
  [Enter]      Accept and save this configuration
  [type text]  Describe changes, then press Enter
  q            Quit without saving

Your choice []:
> Add a drug_interactions task and a diagnosis_support task.

Applying changes via openai/gpt-4o-mini...
[done]  2 tasks added, 1 task updated.

Your choice []:
>  [Enter — accept]

Save to file (or press Enter to print to stdout) []:
> experiments/medical-qa-benchmark.yaml

✔  Config saved to: experiments/medical-qa-benchmark.yaml

Next steps:
  coeval probe --config experiments/medical-qa-benchmark.yaml
  coeval plan  --config experiments/medical-qa-benchmark.yaml
  coeval run   --config experiments/medical-qa-benchmark.yaml
```

Figure 17 (Appendix): CoEval wizard — model selection and pipeline configuration. Teachers, students, and judges are assigned from available provider pools before generating the final YAML.

medical-qa-benchmark.yaml — CoEval Configuration

```yaml
# CoEval — medical-qa-benchmark.yaml   (3 models x 3 roles x 2 tasks x 30 items = ~81 k calls)

# CoEval Experiment Configuration — medical-qa-benchmark
# ==========================================================
# Purpose : Evaluate 3 frontier models on clinical NLP tasks.
# Models  : gpt-4o-mini (OpenAI), claude-3-5-haiku (Anthropic),
#           gemini-2.0-flash (Gemini) — each holds all 3 roles.
# Tasks   : 3 tasks x 30 items = 90 datapoints per teacher.
# Budget  : ~81,000 LLM calls  (3 teachers x 3 tasks x 30 items
#           x 3 students x 3 judges + phases 1-3).
# ==========================================================

# ── MODELS ──────────────────────────────────────────────

models:

  - name: gpt-4o-mini
    interface: openai
    parameters:
      model: gpt-4o-mini
      temperature: 0.7
      max_tokens: 512
    roles: [teacher, student, judge]
    role_parameters:
      teacher: {temperature: 0.0, max_tokens: 512}
      student: {temperature: 0.7, max_tokens: 256}
      judge: {temperature: 0.0, max_tokens: 128}

  - name: claude-haiku
    interface: anthropic
    parameters:
      model: claude-3-5-haiku-20241022
      temperature: 0.7
      max_tokens: 512
    roles: [teacher, student, judge]
    role_parameters:
      teacher: {temperature: 0.0, max_tokens: 512}
      student: {temperature: 0.7, max_tokens: 256}
      judge: {temperature: 0.0, max_tokens: 128}

  - name: gemini-flash
    interface: gemini
    parameters:
      model: gemini-2.0-flash
      temperature: 0.7
      max_tokens: 512
    roles: [teacher, student, judge]
    role_parameters:
      teacher: {temperature: 0.0, max_tokens: 512}
      student: {temperature: 0.7, max_tokens: 256}
      judge: {temperature: 0.0, max_tokens: 128}

# ── TASKS ───────────────────────────────────────────────

tasks:

  - name: clinical_reasoning
    description: >
      Answer clinical questions requiring multi-step diagnostic
      logic and evidence-based medicine.
    output_description: >
      Structured answer: lead diagnosis, key differentials,
      supporting evidence, and management recommendation.
    target_attributes:
      specialty: [cardiology, neurology, oncology, infectious_disease]
      question_type: [diagnosis, treatment_plan, prognosis, differential]
      difficulty: [step1_level, step2_level, attending_level]
      patient_age: [pediatric, adult, geriatric]
    nuanced_attributes:
      evidence_level: [guideline_based, clinical_judgment, controversial]
      presentation: [acute, chronic, incidental_finding]
    sampling:
      target: [1, 2]
      nuance: [1, 1]
      total: 30
    rubric:
      accuracy: "Clinically correct per current evidence-based guidelines."
      reasoning: "Reasoning chain is explicit, step-by-step, and sound."
      completeness: "Key differentials and red flags are addressed."
      safety: "No dangerous advice; contraindications noted."
      clarity: "Answer is well-structured and easy for a clinician to act on."
    evaluation_mode: single

  - name: drug_interactions
    description: >
      Identify and explain clinically significant drug interactions
      for described medication regimens.
    output_description: >
      List of significant interactions, mechanism, severity rating
      (mild/moderate/severe), and recommended management.
    target_attributes:
      n_drugs: [two_drugs, three_drugs, polypharmacy_5plus]
      severity: [mild, moderate, severe, contraindicated]
      drug_class: [cardiovascular, psychotropic, antibiotic, anticoagulant]
      patient_risk: [standard_adult, renal_impairment, hepatic_impairment, elderly]
    nuanced_attributes:
      interaction_type: [pharmacokinetic, pharmacodynamic, both]
      clinical_context: [inpatient, outpatient, emergency]
    sampling:
      target: [1, 2]
      nuance: [1, 1]
      total: 30
    rubric:
      identification: "All clinically significant interactions are identified."
      mechanism: "Pharmacological mechanism is correctly described."
      severity_rating: "Severity is correctly categorised per standard references."
      management: "Management recommendation is specific, safe, and actionable."
      completeness: "No major interactions are omitted."
    evaluation_mode: single

# ── EXPERIMENT ──────────────────────────────────────────

experiment:
  id: medical-qa-benchmark
  storage_folder: Runs
  log_level: INFO
  generation_retries: 2

  phases:
    attribute_mapping: New
    rubric_mapping: New
    data_generation: New
    response_collection: New
    evaluation: New

  quota:
    gpt-4o-mini: {max_calls: 5000}
    claude-haiku: {max_calls: 5000}
    gemini-flash: {max_calls: 5000}
```

Figure 18 (Appendix): Example YAML configuration for the medium-benchmark-v1 experiment. The declarative format specifies task definitions, attribute axes, model assignments, and sampling parameters in a single reproducible artifact.

A.2 Detailed Evaluation Reports

CoEval generates per-judge, per-criterion, and per-model HTML reports from every completed pipeline run. The following figures illustrate the depth of diagnostic information available to practitioners, using the medium-benchmark-v1 experiment as a representative example.

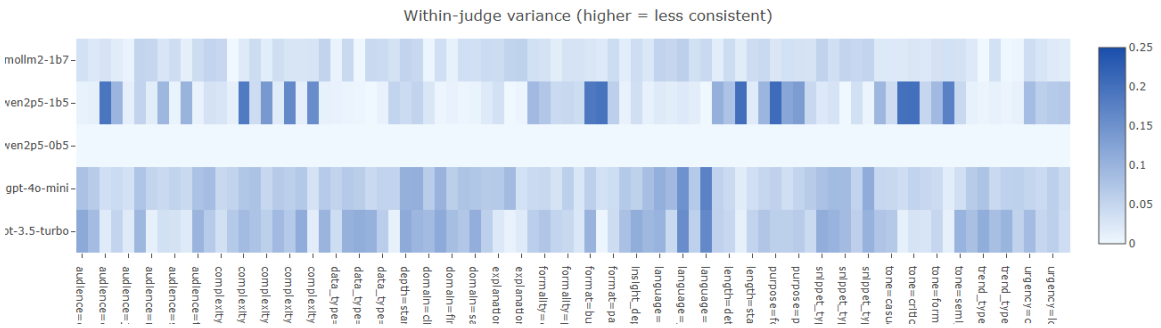
CoEval EEA · Judge Consistency View Experiment: **medium-benchmark-v1** Tasks: 4 Models: 5 Datapoints: 397 ⚠ 1592 self-judging ⚠ 1599 self-teaching Analyzed: 2026-03-01T13:52:59Z

Total eval records: 7987 | Valid: 7978 | Invalid: 9

TASK: All STUDENT: All ASPECT: All

View 1 — Within-Judge Score Variance by Target Attribute (Heatmap)

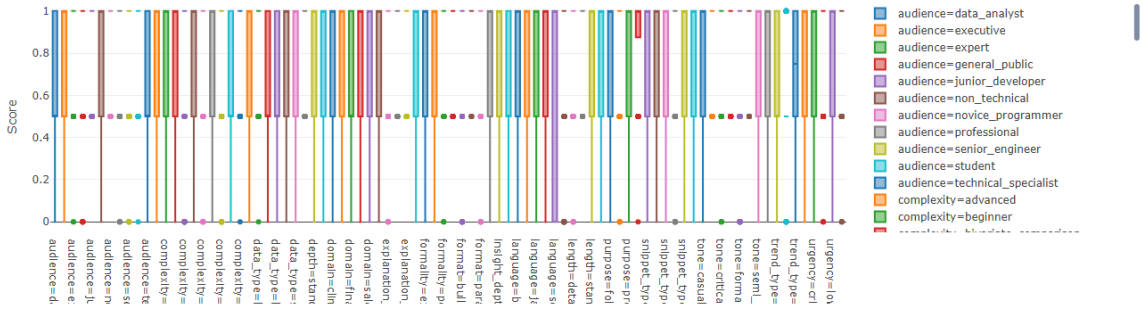
ℹ What does this show?



View 2 — Judge Score Distribution by Target Attribute (Box Plots)

ℹ What does this show?

gpt-3.5-turbo



View 3 — Score Calibration Table

ℹ What does this show?

JUDGE	ASPECT	MEAN	STD DEV	% HIGH	% MEDIUM	% LOW	MOST COMMON	N
gpt-3.5-turbo	accuracy	0.929	0.217	89.1%	7.6%	3.3%	High	996
gpt-3.5-turbo	actionability	0.714	0.368	57.6%	27.6%	14.8%	High	996
gpt-3.5-turbo	appropriate_caveats	0.630	0.298	34.1%	57.9%	8.1%	Medium	496
gpt-3.5-turbo	appropriate_level	0.903	0.270	87.3%	5.9%	6.7%	High	505
gpt-3.5-turbo	appropriate_tone	0.921	0.198	85.4%	13.4%	1.2%	High	500
gpt-3.5-turbo	clarity	0.893	0.226	80.4%	17.8%	1.8%	High	1501
gpt-3.5-turbo	completeness	0.836	0.297	73.8%	19.5%	6.6%	High	1505
gpt-3.5-turbo	conciseness	0.853	0.268	74.6%	21.4%	4%	High	500
gpt-3.5-turbo	insight_quality	0.756	0.300	56.7%	37.9%	5.4%	High	496
gpt-3.5-turbo	practical_value	0.492	0.244	11.1%	76.2%	12.7%	Medium	505
gpt-3.5-turbo	professionalism	0.850	0.232	70.2%	29.6%	0.2%	High	500
gpt-3.5-turbo	readability	0.978	0.140	97.4%	0.8%	1.8%	High	500
gpt-3.5-turbo	statistical_literacy	0.619	0.332	36.7%	50.4%	12.9%	Medium	496
gpt-3.5-turbo	structure	0.900	0.266	86.1%	7.7%	6.1%	High	505
gpt-3.5-turbo	technical_accuracy	0.926	0.227	89.1%	6.9%	4%	High	505
gpt-3.5-turbo	tone_consistency	0.968	0.161	95.8%	2%	2.2%	High	500
gpt-4o-mini	accuracy	0.816	0.301	69.7%	23.8%	6.5%	High	996
gpt-4o-mini	actionability	0.689	0.270	40.7%	56.5%	2.8%	Medium	996
gpt-4o-mini	appropriate_caveats	0.520	0.235	13.1%	77.8%	9.1%	Medium	496
gpt-4o-mini	appropriate_level	0.777	0.335	65.5%	24.4%	10.1%	High	505
gpt-4o-mini	appropriate_tone	0.830	0.265	68.8%	28.4%	2.8%	High	500
gpt-4o-mini	clarity	0.873	0.239	76.7%	21.3%	2%	High	1501
gpt-4o-mini	completeness	0.790	0.281	61.7%	34.8%	3.6%	High	1505
gpt-4o-mini	conciseness	0.569	0.283	23.8%	66.2%	10%	Medium	500
gpt-4o-mini	insight_quality	0.717	0.258	44.4%	54.6%	1%	Medium	496

Figure 5 (Appendix): Example per-judge consistency report (medium-benchmark-v1). SmolLM2-1.7B exhibits near-random pairwise agreement (κ ≈ 0.003) while GPT-4o-mini shows substantially higher reliability.

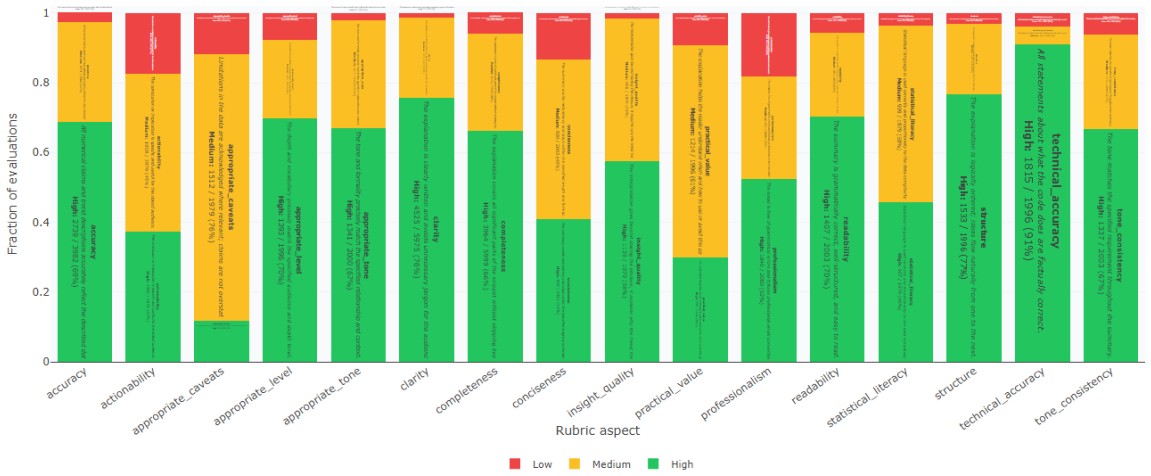
CoEval EEA · Score Distribution Experiment: **medium-benchmark-v1** Tasks: 4 Models: 5 Datapoints: 397 ⚠ 1592 self-judging ⚠ 1599 self-teaching Analyzed: 2026-03-01T13:52:55Z

Total eval records: 7987 | Valid: 7978 | Invalid: 9

TASK: All JUDGE: All TEACHER: All STUDENT: All

Student Score Distribution

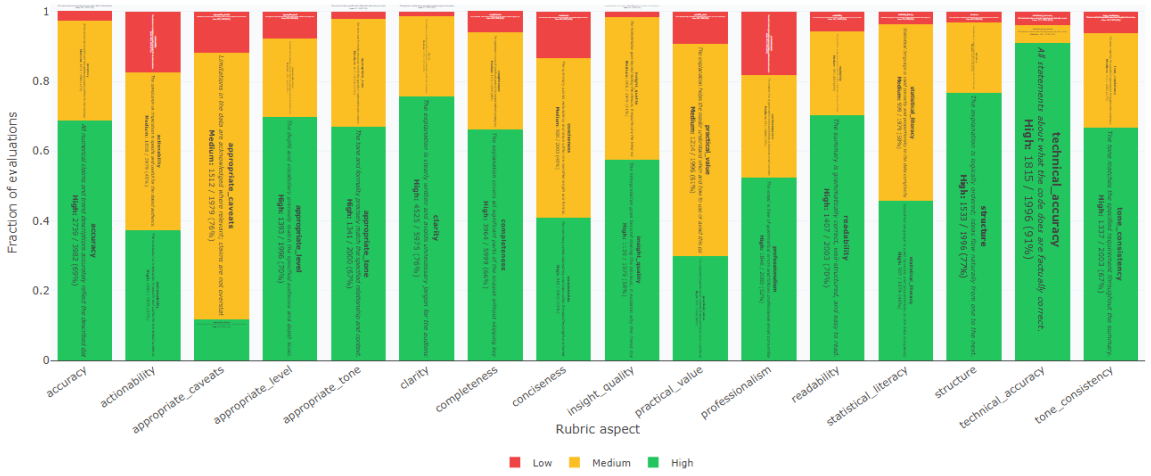
Aggregate by: Rubric aspect View: Stacked (H / M / L)



About this figure

Teacher Score Distribution

Aggregate by: Rubric aspect View: Stacked (H / M / L)



About this figure

Judge Score Distribution

Aggregate by: Rubric aspect View: Average (mean score)

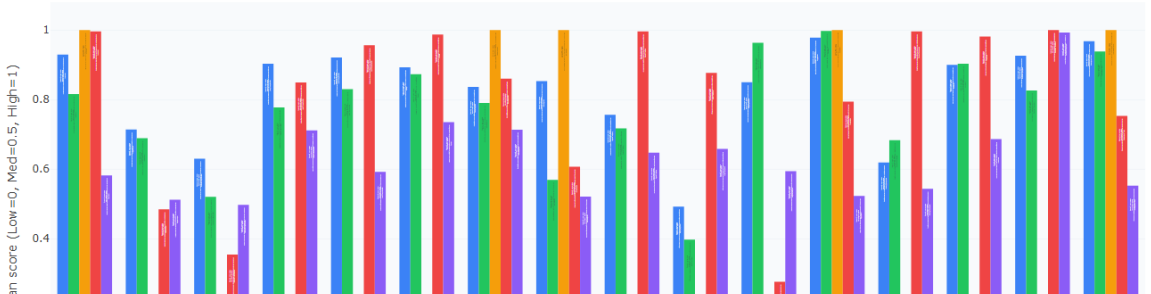


Figure 6 (Appendix): Example score distribution report (medium-benchmark-v1). Data interpretation consistently yields the lowest scores (0.63–0.73); text summarization yields the highest (0.79–0.87).

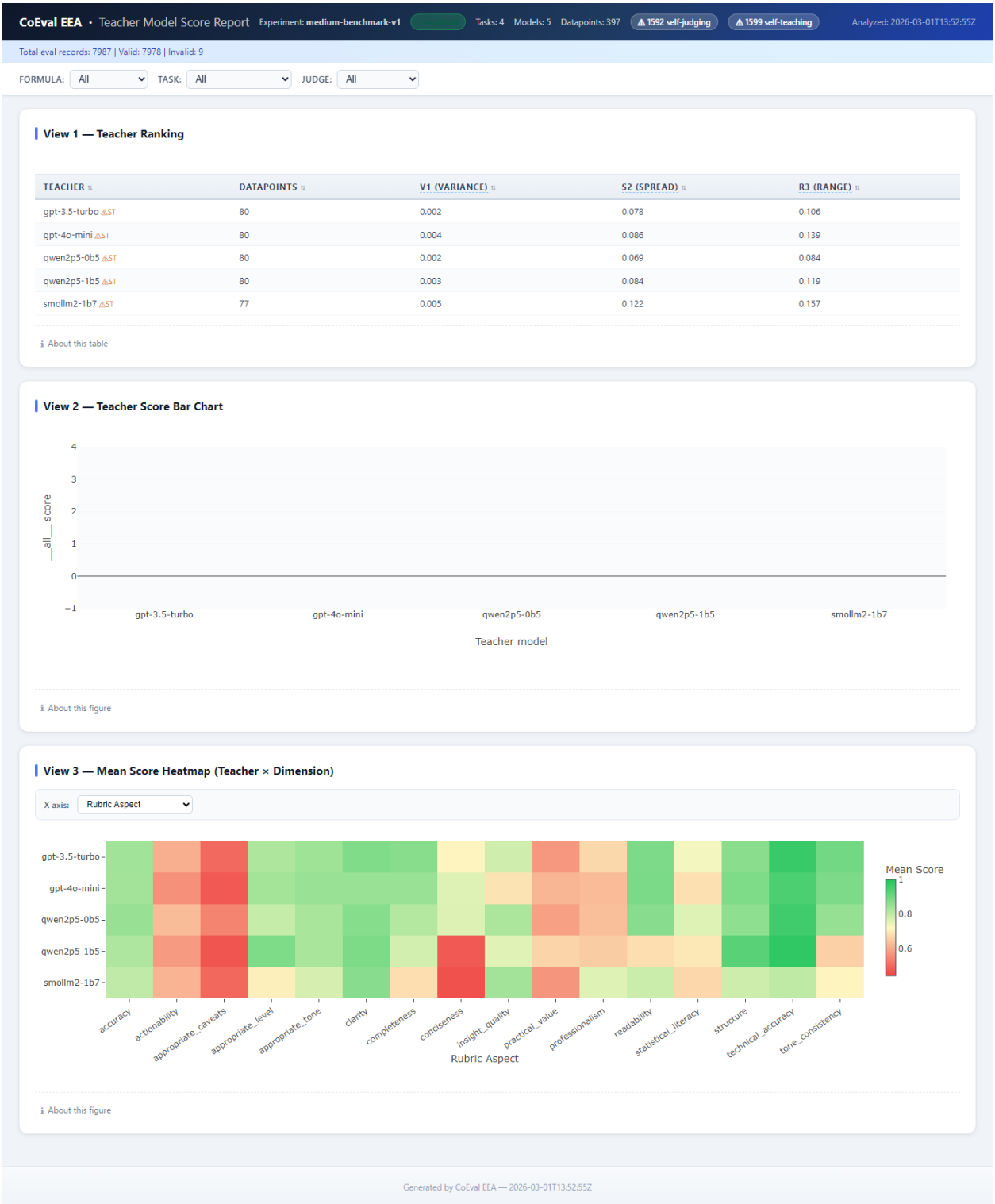


Figure 7 (Appendix): Example teacher discrimination report (medium-benchmark-v1). SmolLM2-1.7B achieves the highest V1 score (0.0046), indicating prompt diversity rather than generation quality drives discrimination power.

Figure 14 (Appendix): Example student performance report (medium-benchmark-v1). GPT-4o-mini achieves the highest mean score (0.832); Qwen2.5-0.5B the lowest (0.648).

CoEval EEA · Coverage Summary Experiment: **medium-benchmark-v1** Tasks: 4 Models: 5 Datapoints: 397 ⚠ 1592 self-judging ⚠ 1599 self-teaching Analyzed: 2026-03-01T13:52:55Z

Total eval records: 7987 | Valid: 7978 | Invalid: 9

Experiment Overview

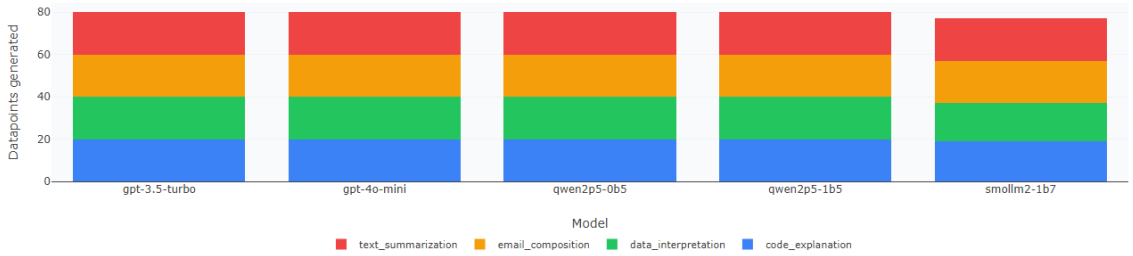
4	5	5	5	397	7987	7978	100%
TASKS	TEACHERS	STUDENTS	JUDGES	DATAPOINTS	TOTAL EVALS	VALID EVALS	COVERAGE

● ✓ Attr Mapping ● ✓ Rubric Mapping ● ✓ Data Gen ● ✓ Resp Collection ● ✓ Evaluation

ℹ About this panel

Teacher Coverage

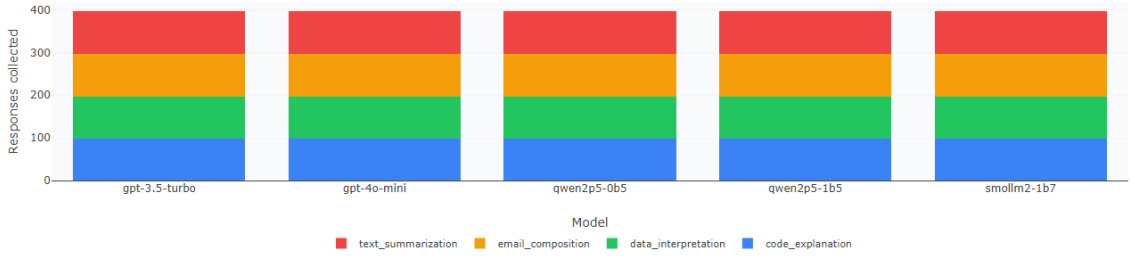
Stack by: Task Group by: Model (default) ⊟ Filter ▾ Bars show datapoints generated; hover for counts.



ℹ About this figure

Student Coverage

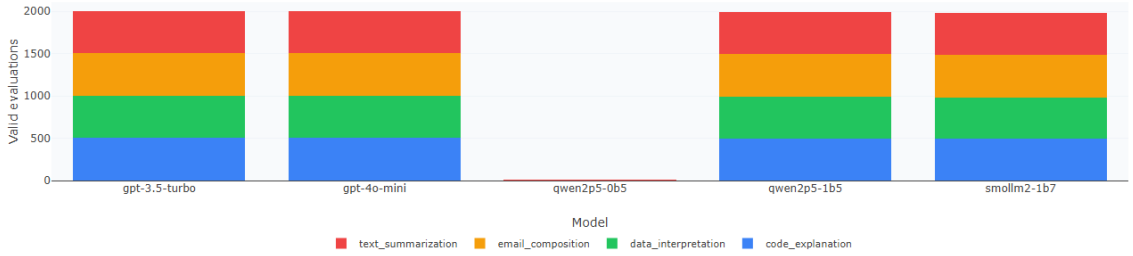
Stack by: Task Group by: Model (default) ⊟ Filter ▾ Bars show responses collected; hover for counts.



ℹ About this figure

Judge Coverage

Stack by: Task Group by: Model (default) ⊟ Filter ▾ Bars show valid evaluations; hover for counts.



ℹ About this figure

Figure 15 (Appendix): Example attribute coverage report (medium-benchmark-v1). All target attribute combinations are covered, validating CoEval's stratified sampling mechanism.

CoEval EEA · Judge Model Score Report Experiment: **medium-benchmark-v1** Tasks: 4 Models: 5 Datapoints: 397 ⚠ 1592 self-judging ⚠ 1599 self-teaching Analyzed: 2026-03-01T13:52:56Z

Total eval records: 7987 | Valid: 7978 | Invalid: 9

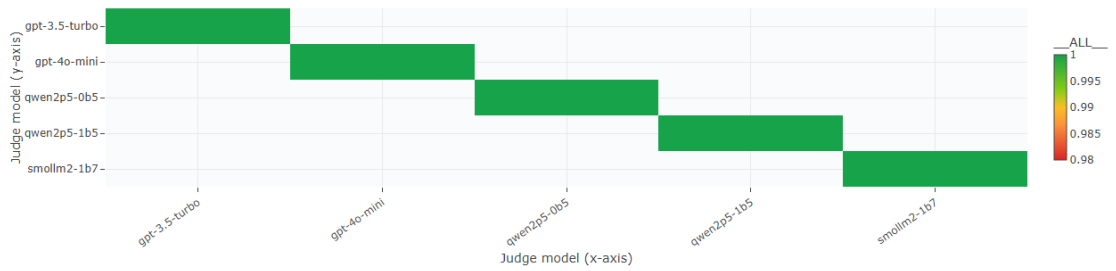
AGREEMENT METRIC: All

View 1 — Judge Ranking

JUDGE	SPA	WPA	K	K LABEL	COVERAGE-WEIGHTED	VALID EVALS
gpt-3.5-turbo ⚠SJ	0.664	0.809	0.137	Slight	0.183	11006
gpt-4o-mini ⚠SJ	0.626	0.792	0.135	Slight	0.181	11006
qwen2p5-0b5 ⚠SJ	0.733	0.867	0.000	Slight	0.000	15
qwen2p5-1b5 ⚠SJ	0.620	0.773	0.087	Slight	0.087	10946
smollm2-1b7 ⚠SJ	0.323	0.653	0.022	Slight	0.030	10892

ℹ About this table

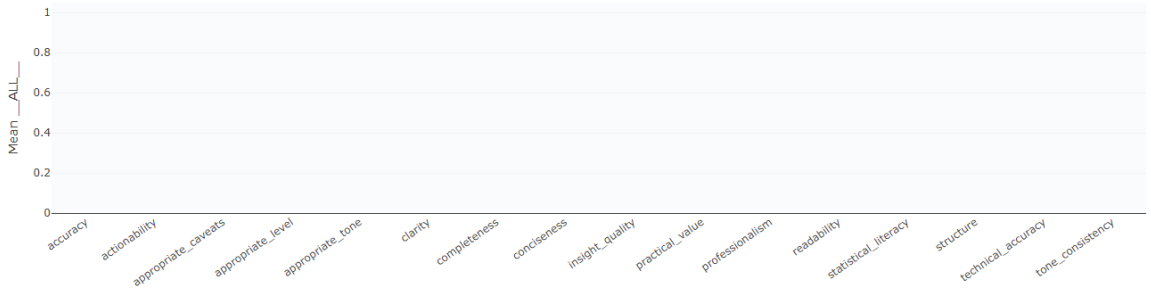
View 2 — Pairwise Agreement Matrix



ℹ About this figure

View 3 — Agreement by Dimension

X axis: Rubric Aspect



ℹ About this figure

Generated by CoEval EEA — 2026-03-01T13:52:56Z

Figure 19 (Appendix): Example detailed judge consistency report (medium-benchmark-v1). Shows SPA, WPA, and κ per judge and per criterion. 'Professionalism' exhibits the lowest cross-judge agreement (SPA = 0.294).

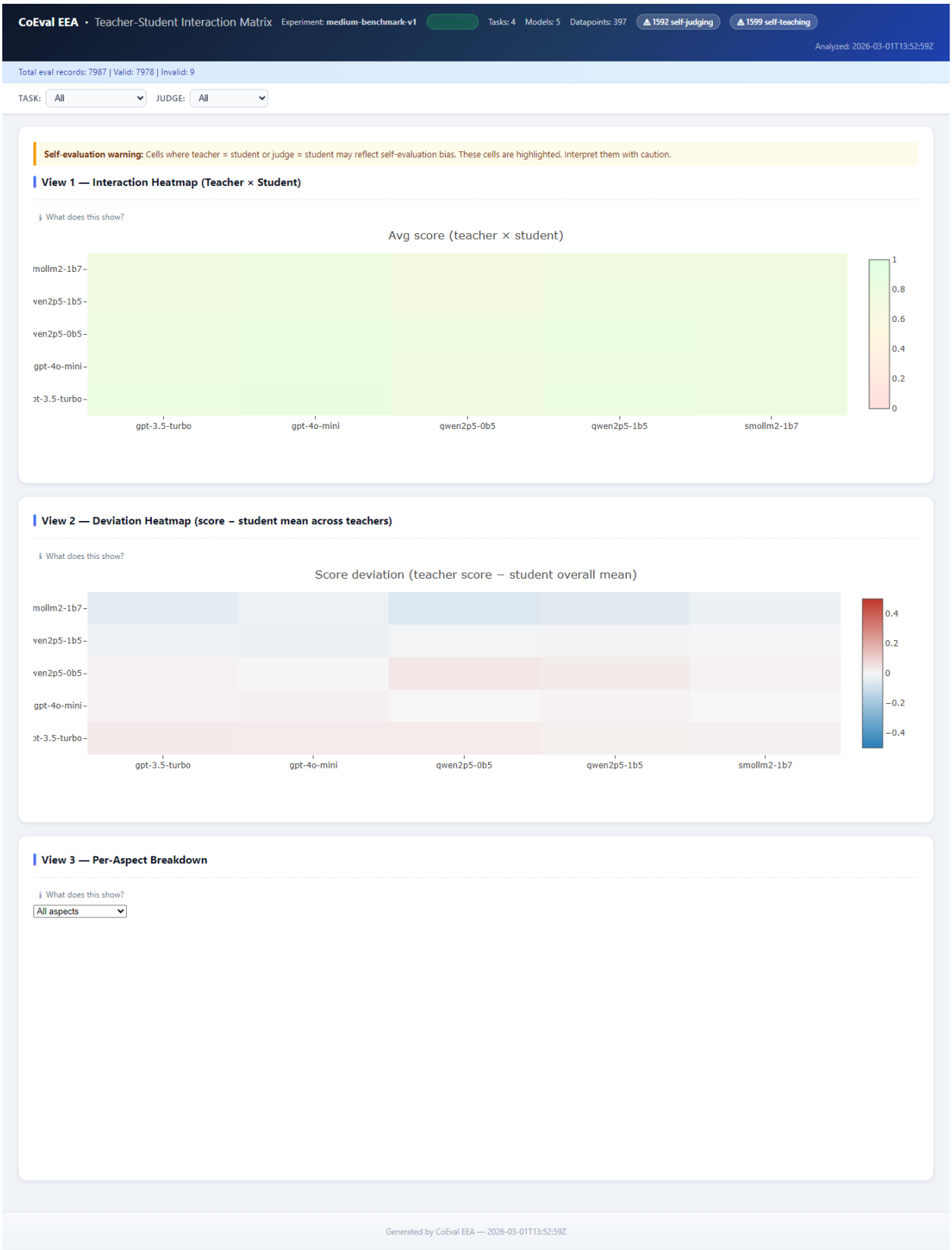


Figure 20 (Appendix): Example judge–criterion interaction matrix (medium-benchmark-v1). Heatmap cells show mean WPA for each (judge, criterion) combination.

CoEval EEA · Coverage Summary Experiment: **medium-benchmark-v1** Tasks: 4 Models: 5 Datapoints: 397 ⚠ 1592 self-judging ⚠ 1599 self-teaching Analyzed: 2026-03-01T13:52:55Z

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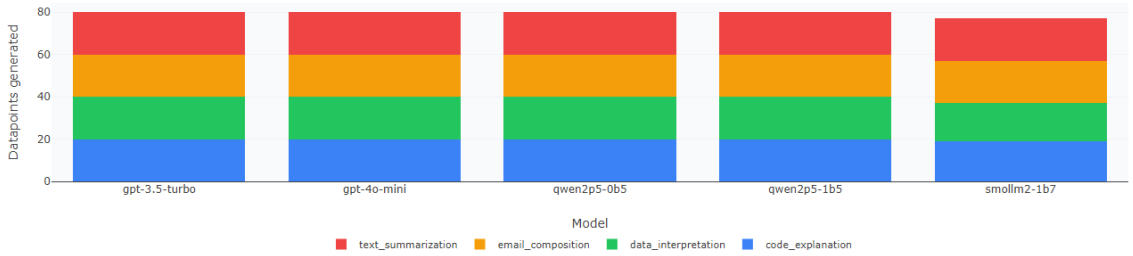
ℹ About this panel

Teacher Coverage

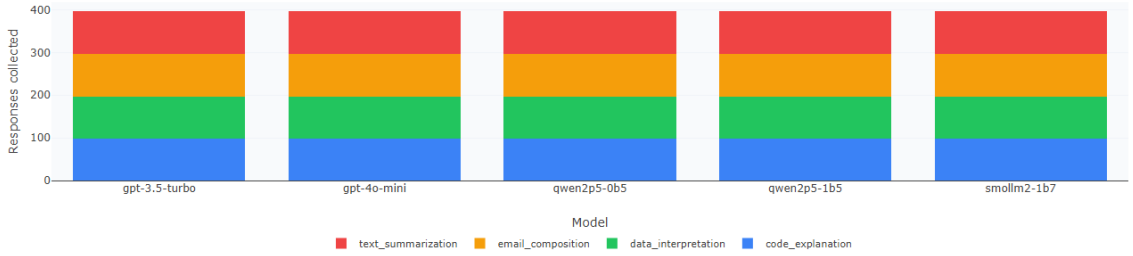
Stack by: Task Group by: Model (default) ⊟ Filter ▾ Bars show datapoints generated; hover for counts.



ℹ About this figure

Student Coverage

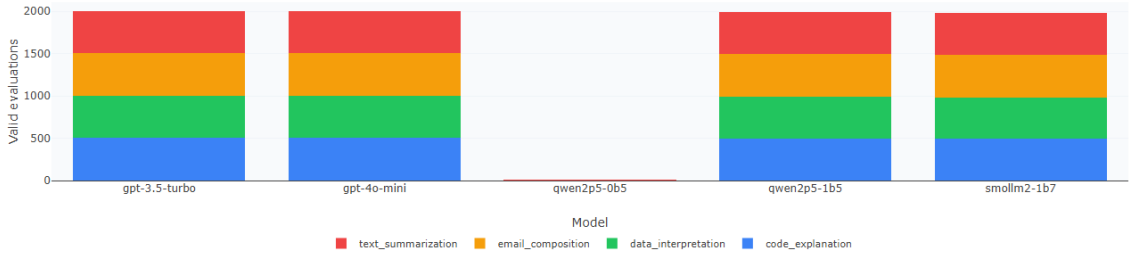
Stack by: Task Group by: Model (default) ⊟ Filter ▾ Bars show responses collected; hover for counts.



ℹ About this figure

Judge Coverage

Stack by: Task Group by: Model (default) ⊟ Filter ▾ Bars show valid evaluations; hover for counts.



ℹ About this figure

Figure 21 (Appendix): Example detailed attribute coverage report (medium-benchmark-v1). Per-cell counts for all target and nuanced attribute combinations.

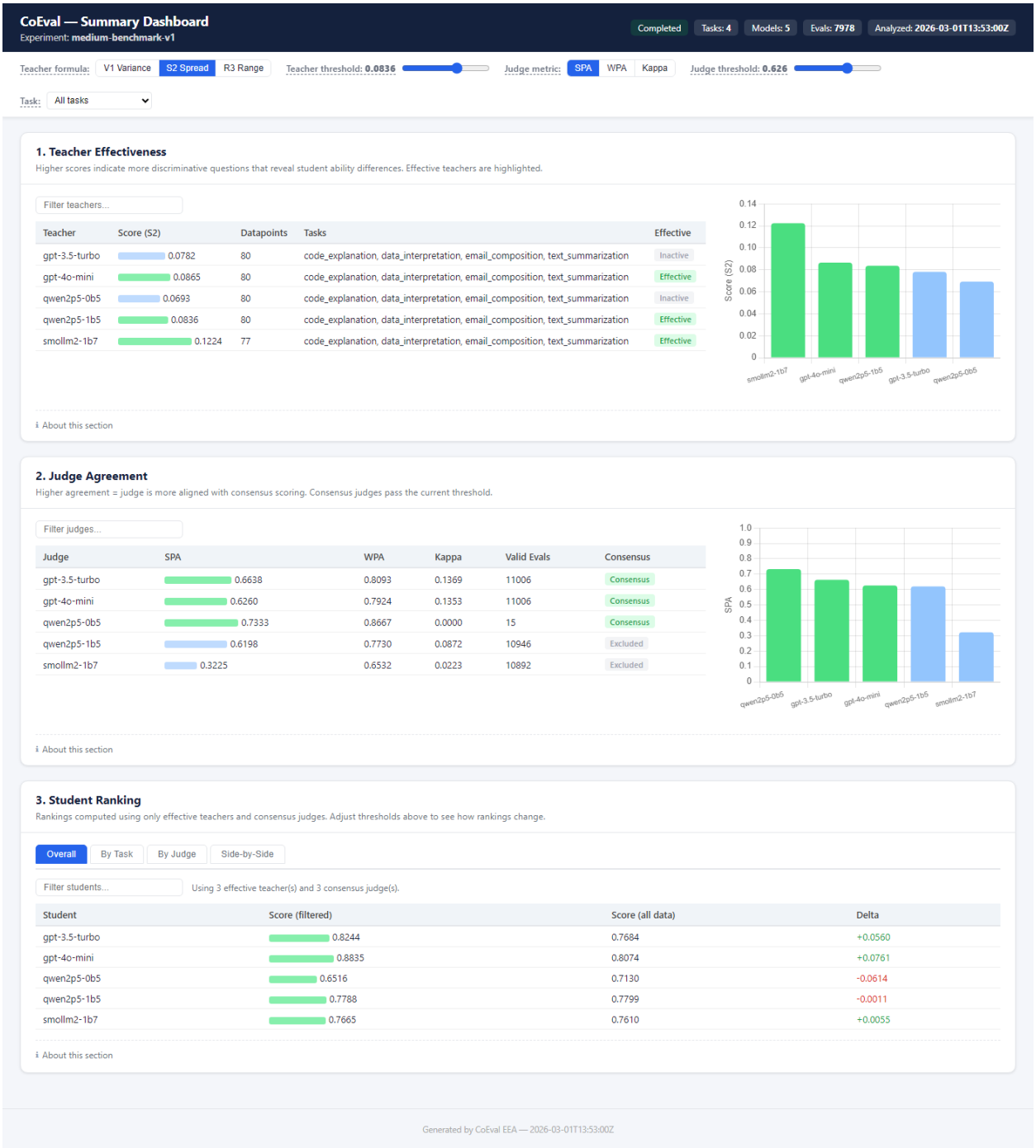


Figure 22 (Appendix): Example full summary report (medium-benchmark-v1). Shows phase completion status, item counts, total cost, and per-model aggregate scores.

Footnotes

[1] Code and data available at: https://github.com/ApartsinProjects/CoEval

[*] Results marked with (*) are projected values — not experimentally measured. These are informed estimates derived from the real medium-benchmark-v1 data and published baselines, pending future validation experiments. See the Disclaimer section for full details.